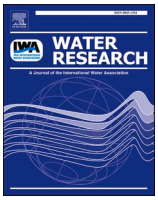
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Enhancing hydrolysis acidification process for coal gasification wastewater treatment through micro-aeration strategy: The role of micro-oxygen environment

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ABSTRACT

Micro-aeration has been regarded as an efficient, economical, and promising strategy for enhancing the anaerobic hydrolysis acidification (HA) of toxic and recalcitrant organics. However, the available information on both its application in actual refractory industrial wastewater and the molecular-level interrelated mechanisms of functional genes and functional enzymes in pollutants degradation is limited. Herein, micro-aeration HA was exploited to investigate the degradation performance of actual coal gasification wastewater under various dissolved oxygen (DO) concentrations. In contrast, the optimal removal efficiencies of chemical oxygen demand, NH_3—N, total organic carbon, and total phenols reached 53.3 %, 14.63 %, 56.5 % and 48.1 %, respectively, as well as biodegradability and detoxification capacity at 0.2–0.3 mg/L of DO. Especially, the micro-oxygen environment demonstrated a proclivity for the removal of phenolic compounds. And the activities of functional enzymes associated with phenolic degradation (phenol hydroxylase, catechol 1,2-dioxygenase and dehydrogenase) were increased. Besides, organics hydrolysis-related, toxic and recalcitrant organics degradation-related, ring cleavage enzyme-related and facultative species under 0.2–0.3 mg/L of DO were more abundant than that in reactor without DO. Finally, this micro-oxygen condition also improved the abundance of genes expressing functional enzymes involved in the metabolic processes of phenolic biochemical degradation (e.g., phenol/toluene 2-monooxygenase, catechol 1,2-dioxygenase, muconate cycloisomerase). This study provides a theoretical basis for the industrial application of micro-aeration strategy for the detoxification pretreatment of refractory industrial wastewater.

1. Introduction

Refractory industrial wastewater (e.g., coal gasification wastewater (CGW)) originates primarily from chemical and energy sectors (Li et al., 2023). CGW contains complex organic pollutants including phenolic compounds, heterocyclics, and polycyclic aromatic hydrocarbons, exhibiting poor biodegradability and acute biotoxicity (Ma et al., 2022). These characteristics not only hinder conventional treatment processes but also pose significant environmental and public health risks, making CGW management a critical challenge in advancing sustainable coal chemical industries (Gui et al., 2019).

Zero-emission treatment for CGW typically involves ammonia/phenol recovery, biochemical treatment, advanced oxidation, membrane concentration and evaporative crystallization (Ji et al., 2016). The anaerobic process served as a linchpin in facilitating the hydrolysis of substantial recalcitrant organic molecules into biodegradable fractions, enhancing biochemical characteristics and mitigating toxicity (Shi et al., 2022). However, CGW toxicity severely inhibits anaerobic bacteria, reducing pollutant removal efficiency (Li et al., 2022b). Conventional anaerobic processes exhibit limited toxicity resistance and low efficiency, necessitating enhanced pretreatment to optimize downstream aerobic treatment. While additives like diverse co-metabolic substrates (methanol/glucose) or functional materials (Fe_3O_4

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Nomenclature

Abbreviation	Definition
AHA	Anaerobic hydrolysis acidification
ATP	Adenosine triphosphate
BOD	Biological oxygen demand
B/C	BOD/COD
CGW	Coal gasification wastewater
COD	Chemical oxygen demand
DHA	Dehydrogenase
DO	Dissolved oxygen
DOM	Dissolved organic matter
DON	Dissolved organic nitrogen
EC_{50}	Median effect concentration
EEM	Excitation-emission-matrix
EPS	Extracellular polymeric substances
F_{max}	Maximum fluorescence intensity
FT-IR	Fourier transformed infrared spectroscopy
GC	Gas chromatography
HA	Hydrolysis acidification or UASB without dissolved oxygen
HRT	Hydraulic retention time
IC	Ion chromatography
IN	Influent
KEGG	Kyoto Encyclopedia of Genes and Genomes
LB-EPS	Loosely bound extracellular polymeric substances
MAHA	Micro-aeration hydrolysis acidification
MAHA-1 reactor	UASB operating at 0–0.1 mg/L dissolved oxygen
MAHA-2 reactor	UASB operating at 0.2–0.3 mg/L dissolved oxygen
MFI	Mean fluorescence intensity
MLSS	Mixed liquor suspended solids
MLVSS	Mixed liquor volatile suspended solids
NIR	Nitrification inhibition rate
OIR	Oxygen uptake rate inhibition rate
ORP	Oxidation-reduction potential
OUR	Oxygen uptake rate
PARAFAC	Parallel factor method
PBS	Phosphate buffered saline
RPKM	Reads Per Kilobase of exon model per Million mapped reads
SEM	Scanning electron microscopy
S-EPS	Soluble extracellular polymeric substances
TB-EPS	Tightly bound extracellular polymeric substances
TN	Total nitrogen
TOC	Total organic carbon
TP	Total phosphorus
Tph	Total phenols
TU	Toxic unit
UASB	Upflow anaerobic sludge blanket
VFAs	Volatile fatty acids
WWTP	Wastewater treatment plant

nanoparticles/graphene/biochar) have been attempted (Li et al., 2022a; Shi et al., 2019; Wang et al., 2010; Zhu et al., 2018), practical implementation remains constrained by high costs and material loss (Li et al., 2024). This underscores an urgent demand for high-efficiency anaerobic enhancement strategies that integrate innovation, scalability, and cost-effectiveness, ensuring robust CGW detoxification as a critical pretreatment process.

Micro-aeration, by introducing limited air or oxygen, enhanced the hydrolysis acidification (HA) of anaerobic system via promoting volatile fatty acids (VFA) production and wastewater detoxification (Fdz-Polanco et al., 2009). This cost-effective strategy has improved anaerobic HA (AHA) efficiency for toxic organics (e.g., 2,4-dinitrophenol, benzene, xylene, aniline, and aldehydes) in petrochemical (Wu et al., 2015), 2-butenal (Song et al., 2019), and resin (Durán et al., 2008) production wastewater. Key evidence includes: (1) refractory organics removal increased from 38.7 % to 82.1 % under 0.2–0.3 mg/L of dissolved oxygen (DO) (Wu et al., 2015); (2) the removal efficiency of phenols was increased from 11.6 % (anaerobic condition) to 80.7 % (micro-oxygen condition, DO: 0.41 mg/L) (Yang et al., 2018); (3) micro-oxygen also enhanced biodegradability of chemical industrial park wastewater as biological oxygen demand/chemical oxygen demand (B/C) ratio rose from 0.46 to 0.61 after 48 h hydrolysis with carriers at filling ratio of 30 % (An et al., 2020). It also alleviated ammonia nitrogen (NH_3—N) inhibition in the anaerobic expanded granular sludge bed (Haroun et al., 2025). However, current research on micro-aeration in refractory wastewater treatment remains limited to specific industrial streams, with insufficient systematic analysis of oxygen dosage thresholds, process synergies, and long-term stability that critical gaps hinder its industrial scalability. Moreover, there is no systematic understanding of microbial ecology (e.g., microbial activity, variations of extracellular polymeric substances (EPS)) during actual industrial wastewater treatment under micro-oxygen environment.

The facilitative mechanisms of micro-oxygen encompassed the incorporation of oxygen in the ring-opening reactions of cyclic organic compounds, the amplification of enzymatic activity, and the proliferation of functionally specialized microbial communities (Zhang et al., 2021). Although existing studies have focused on microbial community, the molecular genetic and physiological functionality of microbes that enhance the degradation of refractory organics remain significantly understudied. Aromatic organic compounds presented a formidable challenge to anaerobic microbial degradation, with ring-opening reactions constituting a critical rate-limiting step (Mu et al., 2022). Traditional activation of ring-opening was sluggish and typically facilitated by iron reduction (Nzila, 2018). In contrast, micro-aeration expedited this activation by incorporating oxygen molecular into the benzene ring via oxygenase, thereby attenuating the stability of the C—C bonds and rendering the organic compounds more susceptible to microbial cleavage (Fuchs, 2008; Varjani and Upasani, 2017; Zhang et al., 2021). Enzymatic agents such as monooxygenases facilitated the insertion of molecular oxygen into monoaromatic hydrocarbons, converting them to phenols, which subsequently underwent further ring-opening reactions in the presence of phenol hydroxylases or oxygenase (Gullotto et al., 2008). Moreover, the micro-oxygen environment was propitious for the amplification of enzymatic activities germane to organics degradation, particularly extracellular functional hydrolases (amylase, protease, cellulase, and oxygenase, etc.) secreted by facultative anaerobic bacteria (Charles et al., 2009; Lim et al., 2014). Micro-aeration enhanced the relative abundance of facultative bacteria. Notably, the phylum *Firmicutes* exhibited a marked increase in relative abundance, particularly within *Longilinea* and *Clostridium*, resulting in a threefold surge in the concentrations of acetic and butyric acids (Song et al., 2019). This phenomenon has been verified across a range of simulated wastewater studies (Zhang et al., 2022). Consequently, the more detailed examination of microbial species, functional genes, and metabolic pathways would be instrumental in elucidating the mechanistic basis and performance attributes of micro-oxygen, thereby fostering its practical implementation and broader adoption in future applications (Bovio et al., 2019).

Herein, the bench-scale UASB reactors with various aeration conditions (0, 0–0.1 and 0.2–0.3 mg/L of DO) were designed for the HA treatment process of CGW in this study. Then, the removal performance of different pollutants, the biodegradability and biotoxicity of CGW

were investigated and evaluated by optimized bioreactors. Specifically, the transform of dissolved organic matter (DOM) under the anaerobic and micro-oxygen environments was analyzed in detail. Subsequently, microbial activity, functional enzyme activity and EPS variation were tested to explore the effect of micro-oxygen environment on microbes in the system. Finally, we delved into the mechanism of micro-oxygen enhanced HA for actual CGW treatment through microbial community structure, functional gene abundance, and metabolic pathways of representative pollutant (phenol).

2. Materials and methods

2.1. Characteristics of wastewater and inoculum

The actual CGW biological treatment process influent was the effluent after the ammonia/phenol recovery process, and the main purpose of this study was to improve the effect of biological treatment. Therefore, the CGW utilized in this study was sourced from the wastewater treatment plant (WWTP) of Henan Yima Coal-to-Gas Co., Ltd, China, having undergone pretreatments via ammonia stripping and phenolic solvent extraction. Exhibiting dark brown in color with pronounced chromacity, the CGW carried strong aromas of coal tar, as depicted in Fig. S1(a). Detailed water quality matrix is provided in Table S1 with COD, BOD, NH_3—N, total nitrogen (TN), and total organic carbon (TOC) values of 3317 mg/L, 1200 mg/L, 263.2 mg/L, 315.4 mg/L, and 1247 mg/L, respectively. To simulate and reveal the actual biochemical treatment process, the influent remained devoid of any additional inorganic elements, nutrients or co-substrate carbon sources. The inoculation sludge, procured from the anaerobic tank of the above WWTP, manifested black color and lacked optimal flocculation, as illustrated in Fig. S1(b). The concentrations of mixed liquor suspended solids (MLSS) and mixed liquor volatile suspended solids (MLVSS) stood at 6.5 g/L and 5.3 g/L, respectively, yielding the MLVSS/MLSS ratio of ~0.82. The associated sludge volume index was 105, indicative of its excellent settling capacity.

2.2. Batch experiments

Micro-aeration hydrolysis acidification (MAHA) batch experiments were performed in blue-capped bottles, in which 150 mL of CGW and 50 mL of sludge with supernatant discarded by centrifugation were added firstly. Anaerobic control group was purged with N_2 for 15 min and the complete removal of DO was verified by litmus blue test paper, then the rubber cover was immediately tightened. The operation of micro-aeration experiment was directly exposed to the air. Then the bottles were placed in a thermostatic shaker (35 °C), and reacted at 150 rpm for 74.5 h. During the experiment, DO and oxidation–reduction potential (ORP) levels were monitored using a DO sensor (MIK-DY2900 meter, Asmik, China) and an ORP instrument (HQ30d, Hach, USA), respectively. And the DO and ORP values were controlled at 0.27 ± 0.06 mg/L and −273~−294 mV, respectively, thereby satisfying the micro-oxygen conditions. Water samples were taken at the given intervals (0, 13, 26.5, 40.5, 50.5, 67 and 74.5 h) and filtered through the membrane (0.45 μm) for the detection of COD, TOC, NH_3—N and Tph concentrations. Long-term batch mode of CGW degradation performance under aerobic and different DO conditions is described in the Text S1, Supporting Information.

2.3. Bioreactor set-up and operation

The anaerobic HA reactor employed in this study was the UASB reactor, featuring an internal diameter of 30 mm, a height of 900 mm, and an effective volume of 0.5 L. CGW was introduced into UASB through the lower inlet via a peristaltic pump, subsequent to DO elimination via N_2 aeration. The effluent was subsequently channeled from the apex of reactor into a collection reservoir, following sedimentation within a reflux tank. Sludge settled at the base of tank was recirculated into the UASB through a reflux pump, operating at a regulated reflux ratio of 1:6, thereby ensuring homogenous sludge suspension within the reactor at an upward flow rate of 0.55 m/h. Thermal stability within the reactor was maintained at 35 °C via a water bath, and the reactor structure and flow path are delineated in Fig. 1 and Fig. S2.

In this study, three parallel UASB groups (HA, MAHA-1 and MAHA-2) were established. The HA reactor served as the anaerobic HA control group, while MAHA-1 and MAHA-2 reactors functioned as the micro-aeration HA test groups, with DO concentrations meticulously regulated at 0–0.1 mg/L and 0.2–0.3 mg/L, respectively. The reactors employed the microporous aerator (with an aperture of 5 μm) situated at its base, connected to an aeration pump, to establish a micro-oxygen environment. This setup was accurately regulated using DO sensor and ORP instrument, while inspections and recordings were carried out at 3–4 h intervals. Given the expansive diameter of the UASB and the rapid upward sludge flow rate, elevated aeration levels were eschewed to mitigate foam production and potential sludge loss, thus precluding the establishment of higher DO concentration in this investigation.

The initial sludge concentration was set at 6 g/L for all reactors, and the hydraulic retention time (HRT) was calibrated at 36 h. Due to the high toxicity of CGW, the influent concentration was incrementally escalated until stabilization was achieved, over a total operational period of 90 d Reactor operation stages and conditions are shown in Table S2. Water samples were subjected to membrane (0.45 μm) filtration for the assessment of general indicators, while the obtained samples during the stable stage underwent comprehensive analysis for biochemical attributes, toxicity levels, and organic compound characteristics. Notably, during sample collection, immediate adjustments to the valve of aeration pump were executed whenever DO and ORP deviated from the set values.

2.4. Analytical methods

2.4.1. Water quality analysis

The concentrations of COD and NH_3—N were determined by

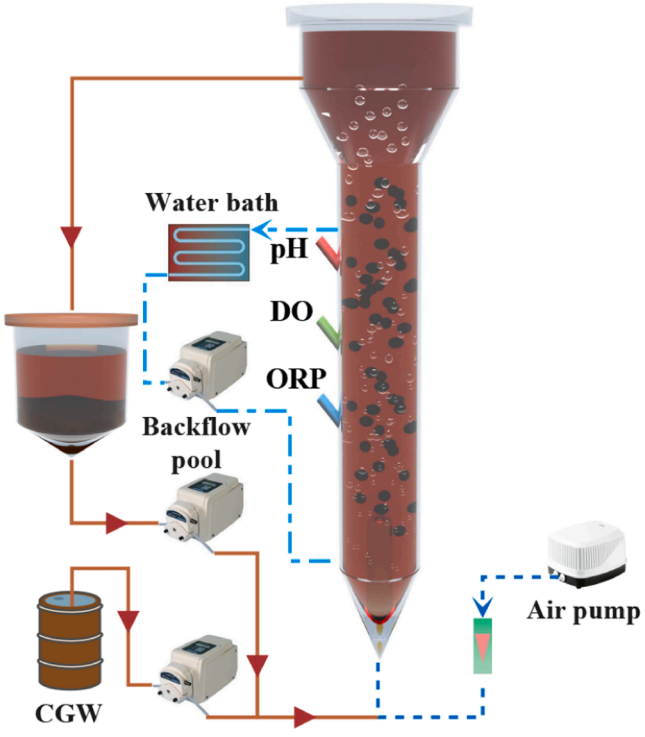


Fig. 1. Experimental set-up of the MAHA process for CGW treatment.

potassium dichromate method and nessler's reagent spectrophotometry (Hei et al., 2022), respectively. Ion chromatography (IC, Thermo-Integrion, US) was utilized for the determination of NO_3^--N and NO_2^--N concentrations. TN was measured by ultraviolet spectrophotometry using basic potassium persulphate, thus the concentration of dissolved organic nitrogen (DON) was equal to the difference between TN concentration and NH_3—N, NO_3^--N and NO_2^--N. Ultraviolet visible spectrophotometer (UV–vis, DR6000, Hash, US) was used for the UV spectroscopy of water samples after 100-fold dilution. Tph and Total phosphorus (TP) were tested by bromine titration and potassium persulfate digestion-molybdenum antimony spectrophotometry, respectively (Guo et al., 2019; Zhang et al., 2023b). TOC was measured using the TOC analyzer (TOC-Lcph, Shimadzu, Japan). MLSS and MLVSS were determined on a gravimetric basis. Gas chromatography (GC, GC-8A, Shimadzu, Japan) was used for the determination of VFAs, and the organic composition of water samples was analyzed by gas chromatography-mass spectrometry (7890A/5975c, GC–MS, Agilent, US). Fourier transformed infrared spectroscopy (FT-IR, IRTracer-100, Shimadzu, Japan) was used to analyze organic functional groups in CGW. Glass electrode method was applied to measure pH (SG2, Hash, US) and conductivity (S40d, Hash, US).

To obtain the composition and characteristics of DOM in CGW, the excitation-emission-matrix (EEM) was measured by a fluorescence spectrophotometer (F-7000, Hitachi, Japan), and the specific operation steps were according to the previous study (Xiao et al., 2016). EEM data was further analyzed by the regional fluorescence integral method. Mean fluorescence intensity (MFI) and different fluorescence ratios (HIXsyn, BIX, FI, HIXem and PeakT/C) were used to characterize the fluorescence properties of organics in water samples according to the description of (Xin et al., 2020). Where, MFI refers to the total fluorescence intensity per unit of TOC, BIX and FI represent the contribution of biologically produced organics, and HIX and PeakT/C are the humification index and biochemical properties, respectively. The parallel factor method (PARAFAC), based on the principle of trilinear decomposition theory and alternating least squares, can identify the characteristic components more clearly than the traditional method of EEM data analysis (Sciscenko et al., 2022). Therefore, the PARAFAC method was further used in this study to investigate the removal of major fluorescent organics from CGW.

2.4.2. Cytotoxicity and specific toxicity tests

Photobacterium luminescens T3 (Beijing Haifuda Technology Co., China) was selected as the test organism for the toxicity analysis of samples, and the toxicity test was carried out according to GB/T 15,441–1995 (China) (Kurvet et al., 2011). The median effect concentration (EC_{50}) was calculated by testing the inhibition rate of water samples at different dilution gradients, then the variation of toxic unit (TU) value of was calculated according to EC_{50} (Eq. (1)).

$$TU = \frac{1}{EC_{50}} \tag{1}$$

The detailed description of total OUR test is provided in Text S2 and Table S3, Supporting Information. The description of nitrification inhibition rate (NIR) is shown in Text S3, Supporting Information.

2.4.3. Microbes activity analysis

The detailed analytical procedure for the live/dead cells of sludge is provided in Text S4, Supporting Information. The adenosine triphosphate (ATP) content of sludge was utilized to characterize the sludge activity at different DO concentrations. The sludge (0.5 mL) was washed three times with 0.1 M phosphate buffered saline (PBS, pH 7.2–7.4) and analyzed by Promega ATP kit. The degradation activity of sludge for organics can be obtained from the dehydrogenase (DHA) activity. Sludge washed with PBS was tested using Mibio Soil Dehydrogenase Kit (96T). Scanning electron microscopy (SEM, PP3010T, Quorum, UK) was applied to observe the surface damage of cell. The sludge sample (0.1 mL) was first diluted 100 times and then filtered through the membrane (0.45 μm). The sludge left on the membrane surface was fixed by glutaraldehyde solution and washed with PBS, followed by dehydration with 50 %, 70 %, 80 %, 90 % and 100 % ethanol and isoamyl acetate for 10 min, and the critical point was dried for 2 h and the surface was sprayed with gold for measurement. The specific detection methods for phenol hydroxylase and catechol dioxygenase are provided in Text S5 and Text S6, Supporting Information, respectively.

2.4.4. EPS extraction and analysis

Thermal extraction was employed to extract EPS from activated sludge. The sludge mixture was washed three times with ultrapure water and centrifuged at 4000 rpm for 5 min. The supernatant was firstly filtered through membrane (0.7 μm), and the remaining sediment was diluted to its original volume with NaCl solution (0.05 %, 50 °C) and then shaken for 1 min and centrifuged at 4000 rpm for 5 min. The supernatant was further filtered through a membrane to obtain loosely bound EPS (LB-EPS). The remaining sediment was again diluted to the original volume with NaCl (0.05 %), centrifuged for 5 min at 4000 rpm after a water bath at 60 °C for 30 min, and the supernatant obtained by filtration was tightly bound EPS (TB-EPS). Phenol-sulfuric acid method was employed to determine the polysaccharide content of LB-EPS and TB-EPS, while protein and humic acid were determined by the modified Lowry method (Zhang et al., 2022). Besides, EEM spectra were further analyzed for LB-EPS and TB-EPS, and the data obtained was processed in conjunction with the PARAFAC technique.

2.4.5. Microbial community and genetic analysis

The detailed description about the operation of microbial community and genetic analysis, including statistical method is shown in Text S7, Supporting Information.

3. Results and discussion

3.1. Performance of bioreactors under different DO concentrations

To ascertain the role of micro-aeration in the HA process of CGW, initial batch experiments were employed to juxtapose the impact of MAHA against anaerobic HA. The anaerobic process exhibits a lag period of ~26 h, during which the concentrations of COD, TOC, NH_3—N, and Tph exhibit a marked decline until 40 h (Fig. S3). Phenolic compounds, particularly polyphenols such as catechol, which are abundant in CGW, exerted pronounced acute toxicity to microbes, conceivably accounting for the protracted lag period in the anaerobic reaction (Kim et al., 2008). Conversely, MAHA exhibits no discernible lag period, signifying that the presence of DO mitigated the wastewater toxicity, thereby abbreviating the lag period of anaerobic process. Anaerobic HA facilitates the removal of 65.6 %, 67.8 %, 17.1 %, and 54.7 % of COD, TOC, NH_3—N, and Tph from CGW, respectively within 74.5 h, while MAHA augments the removal efficiencies of these parameters by 8.6 %, 6.1 %, 4.7 %, and 15.3 %, respectively. Evidently, micro-aeration engenders a substantial enhancement in the removal of COD and Tph through HA process, with Tph removal being particularly noteworthy.

The COD removal efficiency of anaerobic sludge in CGW was also investigated at different DO concentrations (Fig. S4(a)). The results reveal that COD removal efficiency improves from 39.6 % to 59.2 % as the DO concentration increases from 0–0.1 mg/L to 0.8–1 mg/L within 32 d This was because abundant parthenogenetic anaerobic bacteria in the anaerobic sludge could flexibly switch metabolic modes according to the presence or absence of oxygen (Zhang et al., 2012). The aerobic respiration of such species would be accelerated in the presence of DO, thus promoting COD removal. In addition, the effect of biodegradability improvement needs to be emphasized equally. The B/C ratios at DO of 0.2–0.3 mg/L and 0.3–0.5 mg/L increase most significantly from 0.319 (DO: 0–0.1 mg/L) to 0.362 and 0.364, respectively (Fig. S4(b)), suggesting that the appropriate concentration of oxygen contributes to

strengthen the HA, and hydrolyze the toxic macromolecular organics into more small molecules, thus improving the biodegradability of wastewater (Zhang et al., 2021). Nevertheless, the further increase in DO concentration causes a decrease in the B/C ratio, probably because excess oxygen acts as the electron acceptor to intensify aerobic respiration of the parthenogenetic bacteria, consuming excessive amount of degradable micromolecule organics (Wu et al., 2015). Considering the improvement of CGW biodegradation, sludge loss and energy consumption, 0–0.3 mg/L of DO is selected as the condition to examine the micro-oxygen environment of reactor.

The effect of USAB on the removal of pollutants from CGW under different DO conditions was further investigated. In Stage I (1–37 d), all reactors received influent that was diluted and possessed lower concentrations (COD: 1120–1470 mg/L). The effluent concentrations of COD, TOC, NH_3—N, and Tph are ~50 %, ~30 %, ~50 % and ~45 %, respectively (Fig. 2(a-d)). During Stage II (38–51 d), the influent concentration was promptly elevated. This results in the effluent concentration of MAHA-2 gradually descending below that of HA and MAHA-1 reactors, whereas the operation results of HA and MAHA-1 reactors remain similar. This elucidates that the HA process of MAHA-2 is superior in environments with elevated DO levels during the escalation of influent concentration. In Stage III (52–90 d), the influent comprises raw CGW, resulting in gradual differences in pollutant removal among different reactors before stabilization. It is worth noting that the average removal efficiencies of COD, NH_3—N, TOC, and Tph by MAHA-2 stand at 53.3 %, 14.63 %, 56.5 %, and 48.1 %, respectively. Furthermore, the removal efficiencies of COD, TOC, and Tph by MAHA-1 and MAHA-2 surpass those achieved by HA, with MAHA-2 outperforming HA reactor by 14.9 %, 15.5 %, and 24.0 %, respectively. In contrast, the enhancement of Tph removal is most pronounced under micro-oxygen environment. NH_3—N removal efficiency through HA reactor (10.14 %) is relatively low, while MAHA-1 and MAHA-2 improve it by 1.65 % and 4.49 %, respectively. Nitrogen balance analysis during Stage III reveals that the DON concentration in CGW is 39.32

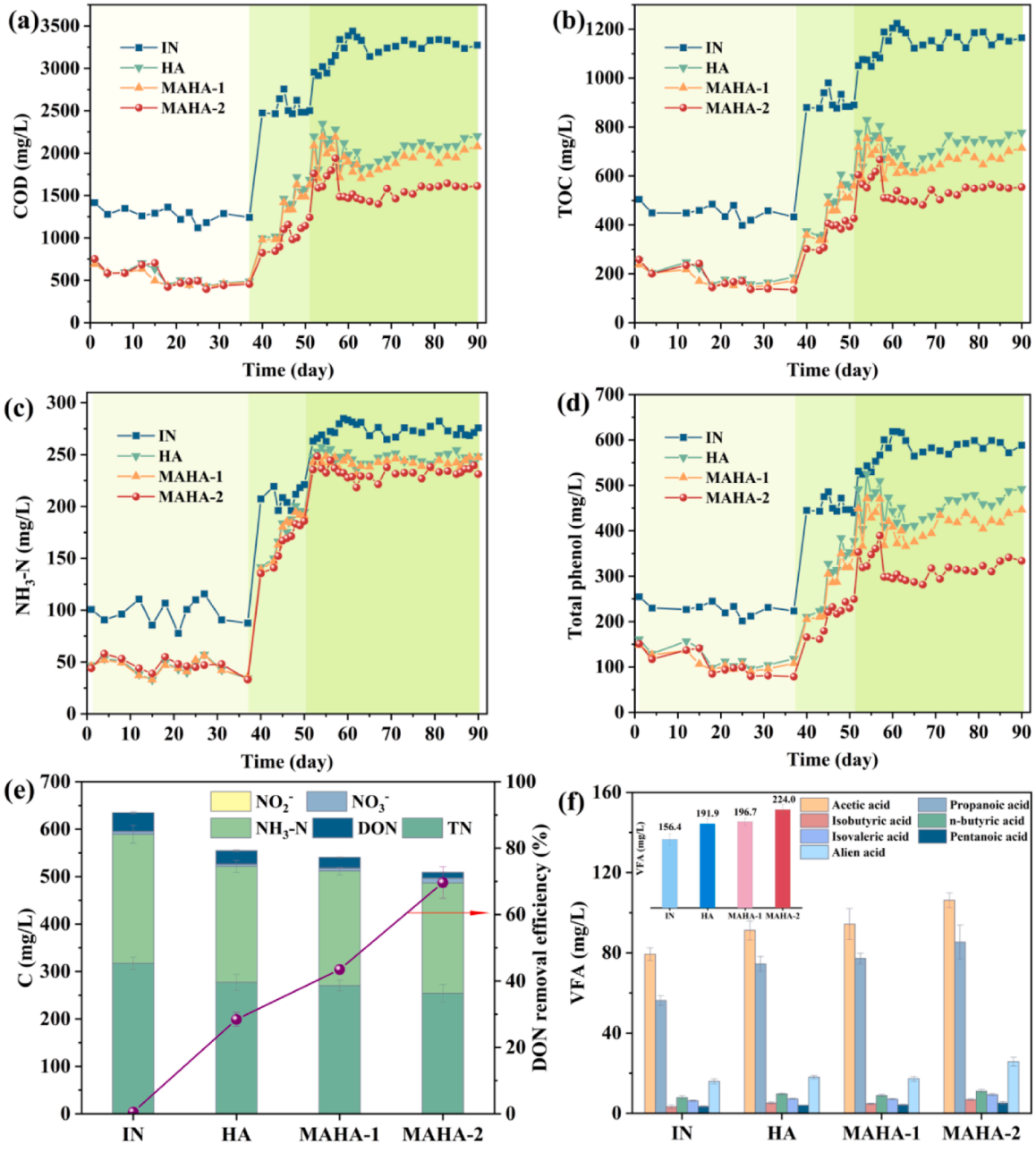


Fig. 2. (a) COD, (b) NH_3—N, (c) TOC and (d) Tph, and the concentrations of (e) various nitrogen species and (f) VFAs under different DO concentrations.

mg/L, predominantly comprising nitrogen heterocyclic organics (e.g., quinoline). Extracellular hydrolases secreted by parthenogenic bacteria hydrolyzed DON to release NH_3—N, thereby elevating its concentration, possibly accounting for the lower NH_3—N removal during HA process (Wang et al., 2020). DON removal efficiencies in HA, MAHA-1, and MAHA-2 reactors are recorded at 28.4 %, 43.4 %, and 69.3 %, respectively (Fig. 2(e)). Consequently, DON removal in MAHA-2 is ~40 % higher than that in HA reactor, probably due to a slight improvement in the NH_3—N removal in MAHA-2.

Fig. S5 lists the DO, ORP and pH values during the daily monitoring of reactor operations. DO levels in MAHA-1 and MAHA-2 exhibit oscillations within 0–0.1 mg/L and 0.2–0.3 mg/L, respectively, while HA reactor diligently upholds a rigorously anaerobic environment (Fig. S5 (a)). In the HA reactor, the ORP registers in the range of −320~−305 mV, whereas MAHA-1 and MAHA-2 maintain ORP values within −300~−281 and −255~−240 mV, respectively, under varying degrees of micro-aeration (Fig. S5(b)). CGW is weakly alkaline and pH values in all reactors are maintained between 7.58 and 7.86 (Fig. S5(c)). Overall, the marginally lower pH value observed in the MAHA-2 reactor can be attributed to the enhanced HA process, which generates a greater volume of VFAs.

Refractory organics underwent an initial hydrolysis, resulting in the generation of intermediates and molecular organics through ring-opening and chain-breaking mechanisms (Chen et al., 2023, 2020). Subsequently, small-molecule VFAs were produced through the activities of fermentation bacteria. During the transportation and preservation of CGW, there was a possibility of partial AHA occurring, thereby resulting in markedly higher concentrations of VFAs (acetic acid, propionic acid, isobutyric acid, *n*-butyric acid, isovaleric acid, pentanoic acid, and alien acid) (Song et al., 2019). During the HA process, the concentrations of acetic acid and propionic acid increase from 79.3 to 56.2 to 91.2 and 74.5 mg/L, respectively, underscoring the role of AHA in acid production (Fig. 2(f)). Though there is minimal variation in acetic acid and propionic acid concentrations in MAHA-1, the effluent of MAHA-2 registers notably higher levels of acetic acid (118.9 mg/L) and propionic acid (94.16 mg/L) in comparison to HA. Moreover, the cumulative VFAs concentrations in the effluents of HA, MAHA-1, and MAHA-2 reactors stand at 184.2, 189.9, and 221.8 mg/L, respectively, signifying that 0.2–0.3 mg/L of DO notably facilitates the acidification process, yielding a greater abundance of small-molecule acids.

3.2. Assessment of the detoxification and DOM removal efficacy

To ensure stable and efficient operation of the subsequent aerobic process, the detoxification of CGW and improvement of biochemical properties were the main objectives of anaerobic pretreatment process (Hei et al., 2022). B/C ratio of CGW is 0.354, which belongs to industrial wastewater that is difficult to be treated biologically (Fig. 3(a)). Subsequently, B/C ratio is improved to 0.368, 0.362 and 0.397 after the treatment of HA, MAHA-1 and MAHA-2 reactors, respectively. It can be observed that MAHA-1 is not effective in enhancing B/C ratio, while MAHA-2 clearly optimizes the biochemical properties of CGW. Initial TU value stands at 38.2, categorizing it as exhibiting high acute toxicity based on toxicity grading standards (Fig. 3(b), Fig. S6(a)). Although the effluent toxicity of all reactors continues to manifest high acute toxicity, the TU value of MAHA-2 (~10) is lower than that of HA reactor, signifying that the reactor with elevated DO levels achieves superior detoxification.

OUR inhibition rate (OIR) was employed to further investigate the improvement of wastewater biodegradability (Fig. S6(b)). OIR value diminishes from initial 89.12 % to 73.61 %, 75.73 %, and 56.42 % after the treatment of HA, MAHA-1, and MAHA-2 reactors, respectively (Fig. 3(c)). This indicates that HA and MAHA-1 reactors exert no conspicuous influence on the biochemical characteristics. Conversely, MAHA-2 markedly elevates the wastewater biodegradability, a phenomenon likely linked to its promotion of HA, resulting in the generation of greater quantity of small-molecule organics and VFAs. Nitrifying bacteria demonstrate sensitivity to wastewater toxicity, and CGW contains a substantial concentration of NH_3—N. To ensure the effective elimination of NH_3—N from the aerobic unit, it becomes imperative to reduce the wastewater toxicity toward nitrifying bacteria. CGW proves lethal to nitrifying bacteria due to its elevated inhibition rate. However,

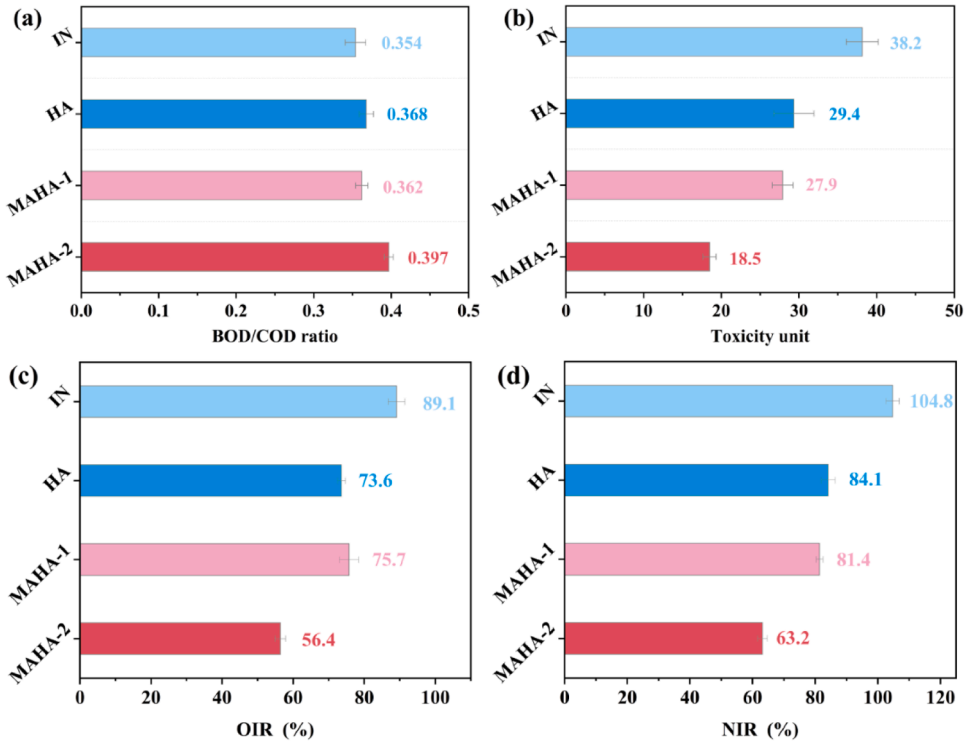


Fig. 3. Variations of (a) B/C ratio, (b) TU, (c) OIR and (d) NIR under different DO concentrations after treatment.

the inhibition of nitrifying bacteria appears to diminish post-treatment with HA (84.3 %), MAHA-1 (81.1 %), and MAHA-2 (63.6 %) reactors, suggesting that HA process could reduce the inhibition of nitrifying bacteria (Fig. 3(d)). Evidently, the NIR of MAHA-2 is lower than that of HA reactor.

The anaerobic process possesses the efficacy of converting high-molecular-weight and toxic organic compounds into smaller and lower toxic substances (Shi et al., 2019). After the direct aerobic degradation of raw CGW (38 h), the COD value remains elevated at 1280 mg/L (Fig. S7(a)). In contrast, the effluent COD values are significantly reduced to 648.2, 622.3, and 465.1 mg/L, respectively, for HA, MAHA-1, and MAHA-2 reactors under the aerobic environment. Meanwhile, MAHA-2 reactor also exhibits better removal efficiency for NH_3—N (Fig. S7(b)). This suggests that the MAHA-2 effluent is more suitable for subsequent aerobic biodegradation, indicative of its lower biotoxicity.

Characteristic organic functional groups present in the wastewater manifest distinctive absorption bands in UV–vis spectroscopy, thereby delineating the attributes of organic pollutants (e.g., aromaticity) (Yang et al., 2024). Noticeably prominent absorption peaks within 180–220 nm signify the prevalence of aromatic compounds, with specific reference to the E1 band (λ_{max}=184 nm) and E2 band (λ_{max}=204 nm) of benzene (Fig. 4(a)) (Wang et al., 2022). It is noteworthy that these bands exhibit a redshift. This spectral shift showed the presence of aromatic organics in the wastewater with larger conjugated systems than benzene ring, implying the attachment of electron-donating substituents to the benzene ring. Subsequent treatment in the reactors results in a conspicuous blueshift of the characteristic peaks at 180–220 nm. This observation indicated that HA process led to a reduction in C═C and π-π conjugated systems or organic compounds containing heteroatoms, and a decline in aromaticity. Moreover, the absorption peaks in the effluents of all reactors exhibited a diminishment compared to the influent, further underscoring the capacity of HA process to reduce these organics. The extent of peak reduction in the absorption bands at 180–220 nm is similar for HA and MAHA-1 reactors, whereas MAHA-2 exhibits a significantly more pronounced reduction in peak intensity. This signifies that MAHA-2 exhibits a stronger efficacy in removing relevant organics from CGW.

UV_{254} is typically employed to elucidate the presence of unsaturated organic compounds with C═C structures, including aromatic compounds. The UV_{254} value of influent is 1.84, and the UV_{254} values treated with HA (1.65), MAHA-1 (1.51) and MAHA-2 (1.06) reactors are reduced to different degrees. It reflects that MAHA-2 effluent contains lower concentration of unsaturated C═C and aromatic organics. Furthermore, UV_{300}/UV_{400} ratio serves as a metric for characterizing the humification of wastewater, with lower value indicating a higher humification and the presence of more benzene-ring organics (Artinger et al., 2000). The UV_{300}/UV_{400} ratio of MAHA-2 effluent is markedly higher than that of influent, HA effluent and MAHA-1 effluent, suggesting lower humification and lower concentration of benzene-ring organics in MAHA-2 effluent.

To investigate the removal characteristics of organic functional groups in CGW by reactors, the FT-IR analysis of influent and effluent was conducted (Fig. 4(b)). The peak at 1588 cm^{-1} corresponds to the skeletal stretching vibration of $\sigma_{C=C}$ aromatic rings, indicative of the notable presence of benzene-ring structures (Radji et al., 2022). The peaks at ~1342 cm^{-1} represent the stretching vibration of C—O bond and the bending vibration of O—H, showing the characteristic of phenolic absorption bands (Yang et al., 2015). The absorption peaks at 1135 and 1031 cm^{-1} denote C—O bonds, while ~831, 766, 695, 656,

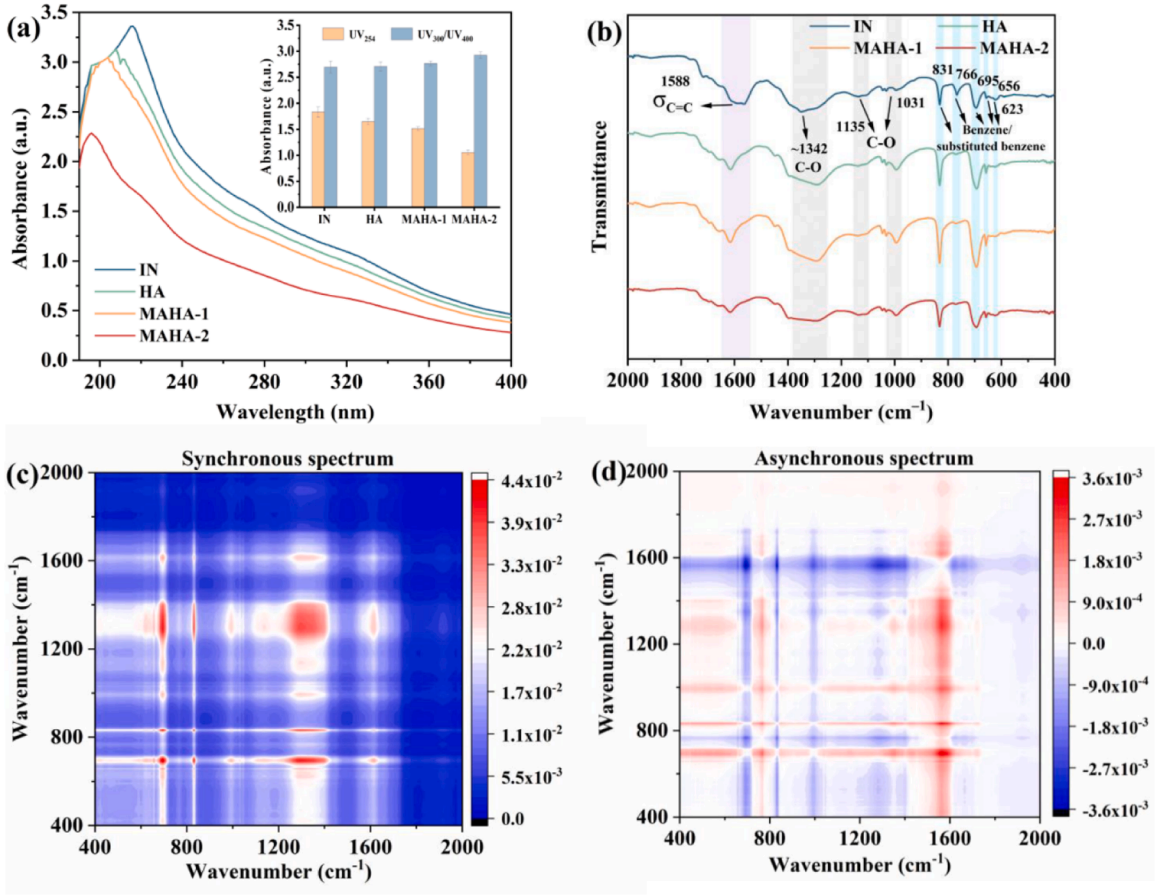


Fig. 4. (a) UV–vis absorption spectra and (b) FT-IR spectra of CGW under different DO concentrations after treatment; The generation of 2D-cos maps ((c) synchronous and (d) asynchronous) from the corresponding FT-IR spectra.

and 623 cm^{-1} signify benzene or substituted benzene, with 766 and 831 cm^{-1} suggesting the disubstituted benzene (Yang et al., 2015). A distinct blueshift is observed at ~1588 cm^{-1} in the effluent, indicating a reduction in π-π conjugated systems due to HA process, leading to a decrease in conjugated functional groups (e.g., benzene rings) (Meng et al., 2016). This further elucidates a significant reduction in benzene-ring organic compounds. Furthermore, the absorbance reduction is observed in the characteristic peaks of effluent across all reactors, with MAHA-2 demonstrating a pronounced decrease in absorbance. This underscores the superior removal efficacy of MAHA-2 towards various organic functional groups in the wastewater.

The interrelations between various organic functional groups were explored by 2D-cos analysis. In synchronous fluorescence spectra, the conspicuous autocorrelation peaks manifest at 695, ~766, ~831, ~1031, ~1342, and ~1588 cm^{-1} (Fig. 4(c)). Among them, the characteristic absorption peak of phenolic hydroxyl (1342 cm^{-1}), monosubstituted benzene (695 cm^{-1}) and disubstituted benzene (831 cm^{-1}) exhibit the highest sensitivity to DO variations (Liu et al., 2016). Additionally, all peaks exhibited orthogonal cross-peaks, signifying that with the DO concentrations changed, the functional groups corresponding to the peaks changed positively in the same direction. No autocorrelation peaks appear in the asynchronous fluorescence spectra and all peaks are cross peaks (Fig. 4(d)). Based on the positive and negative cross peaks, we can determine the variation in the position of characteristic peaks at different DO concentrations. The positive and negative cross peaks represent the order of precedence: ~1588>1342=766>831=695=1031 cm^{-1}. The findings elucidate that the presence of O_2 prioritizes the C═C bond transformation on the benzene ring, followed by the phenolic hydroxyl and disubstituted groups, and ultimately the benzene ring or C—O monosubstituted groups. Based on the aforementioned experimental findings, MAHA-1 does not significantly stimulate DOM compared to HA reactor. Conversely, MAHA-2 exerts a marked influence on DOM removal.

Consequently, the focus was directed towards a comparative analysis of compositional disparities in the DOM of effluent between HA and MAHA-2 reactors by GC–MS. CGW mainly contains phenolic compounds (60.54 %), heterocyclic compounds (21.32 %), long-chain alkanes (3.82 %), and polycyclic aromatic hydrocarbons (3.11 %) (Table S4). Among the phenolic substances, phenol accounts for 28.74 %, accompanied by polyphenols such as methylphenol and catechol, thus indicating that CGW is a typical phenolic wastewater. A marked reduction in phenolic compounds is evident in HA and MAHA-2, with their respective proportions declining to 31.96 % and 21.35 %. The application of microaeration confers upon HA a preferential elimination capacity for phenolic substances, thereby reducing the relative abundance of phenolics (Fig. S8). Concurrently, their respective proportions (heterocyclic compounds, long-chain alkanes, and polycyclic aromatic hydrocarbons) in the effluent of HA reactor stand at 35.98 %, 11.13 %, and 6.55 %, while MAHA-2 reactor registers proportions of 32.82 %, 19.01 %, and 12.2 %. The cumulative proportions in the effluent exceed those in the

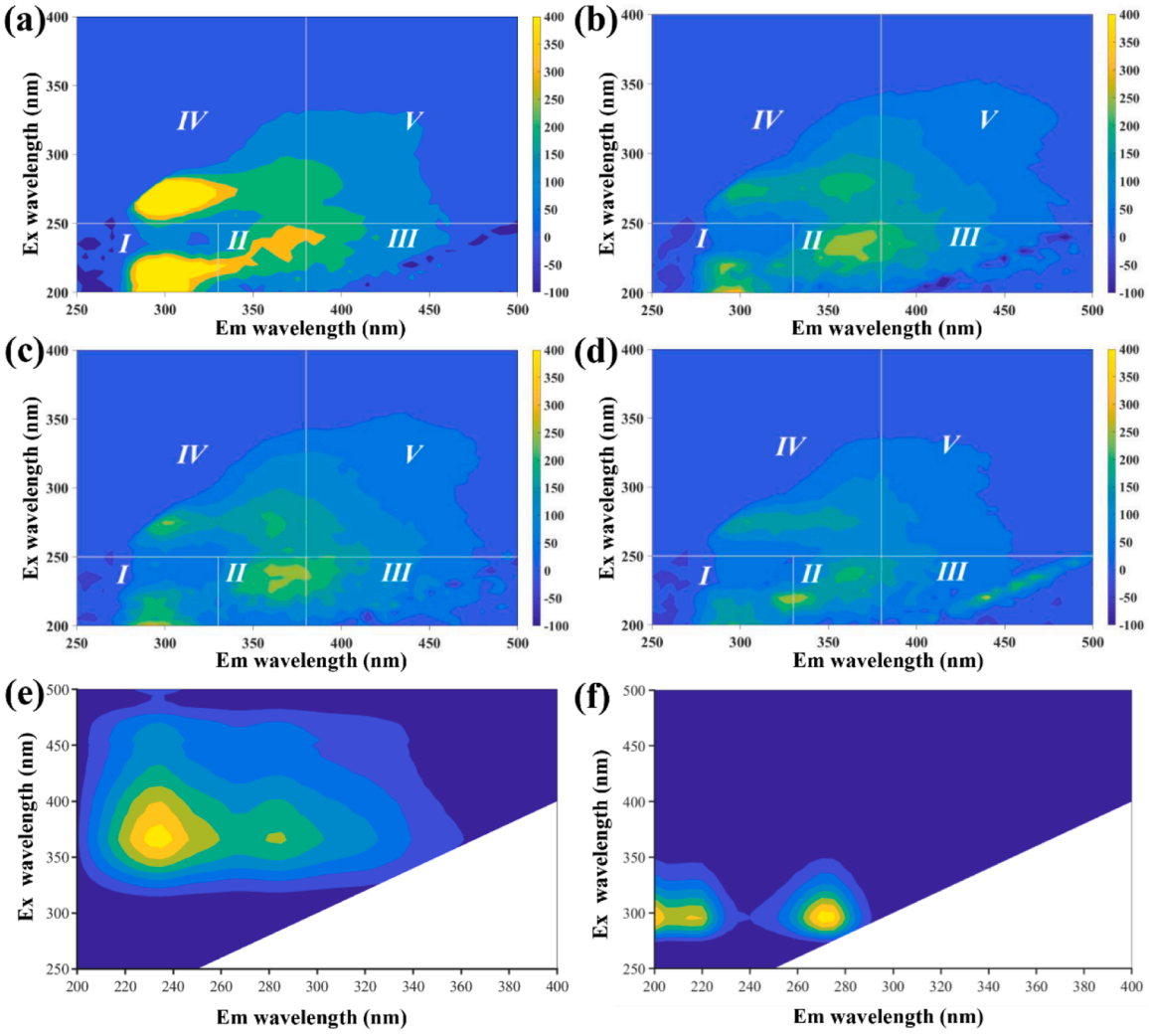


Fig. 5. EEM of (a) raw CGW and samples after the treatment of (b) HA, (c) MAHA-1 and (d) MAHA-2 reactors; The fluorescence spectra of (e) component 1 and (f) component 2.

influent, implying that these species are more recalcitrant and challenging to degrade and remove than phenolic compounds. Additionally, the micro-oxygen environment demonstrates a proclivity for the removal of phenolic compounds over heterocyclic compounds, long-chain alkanes, and polycyclic aromatic hydrocarbons.

The fluorescence spectra of CGW are mainly distributed in regions I, II and IV, denoting that the organic contaminants primarily consist of aromatic proteins and microbial metabolites (Fig. 5(a)). Aromatic proteins generally refer to amino acids with aromatic properties (e.g., tryptophan, tyrosine, etc.), while microbial metabolites can also represent tryptophan, tyrosine or their residues (Rodríguez-Vidal et al., 2022). The distribution of CGW fluorescence peaks is highly similar to the EEM characteristics of phenols (Ex/Em 272/300 nm), toluene, ethylbenzene, and xylene (Ex/Em 210/280 nm, 260/285 nm), with some overlap evident in the characteristic fluorescence peaks of polycyclic aromatic hydrocarbons (e.g., naphthalene: Ex/Em 280/340 nm) (Yu et al., 2025). This observation suggests that the predominant organics are mono-phenylene ring (e.g., phenols) and polycyclic organic substances. There is a discernible reduction in fluorescent constituents in regions I, II, and IV after treatment in all reactors (Fig. 5(b-d)). Notably, the most substantial reduction occurs in microbial metabolites in region IV, which coincides with characteristic absorption peaks associated with benzene rings and phenolic compounds (Fig. S9(a)). This result underscores the superior removal efficacy of HA process for monocyclic organic substances. Additionally, aromatic-like proteins within regions I and II also appear to be reduced in abundance. It is noteworthy that HA and MAHA-1 reactors exhibit comparable effects in the removal of fluorescent components from CGW, whereas MAHA-2 demonstrates the most pronounced efficacy in this regard. This underscores the positive impact of higher DO levels on the enhancement of fluorescent DOM in CGW.

PARAFAC method was adopted to calculate the two main characteristic components of CGW, of which component 1 at Ex/Em of 235/375 nm fluorescence peak is aromatic-like protein II (Fig. 5(e)), and component 2 corresponds to aromatic-like protein I (Yang et al., 2015) (Fig. 5(f)). CGW mainly contains component 2, thus proving again that DOM in wastewater is dominated by monobenzene or phenolic organics. There is a conspicuous reduction in the maximum fluorescence intensity (F_{max}) of component 2 in all reactors. It has substantially decreased from 1441.7 to 703.5, 660.3 and 418.9 au, with the reduction efficiencies of 51.2 %, 54.2 % and 70.9 %, respectively (Fig. S9(b)). Notably, MAHA-2 reactor exhibits an enhancement of ~20 % in reduction efficiency, suggesting that the reactor notably promotes the removal of aromatic-like protein I, monocyclic aromatic compounds, and phenolic organic substances under 0.2–0.3 mg/L of DO. Conversely, component 1 exhibits relatively lower fluorescence intensity in CGW and a less pronounced reduction effect compared to component 2, with reduction efficiencies of 20.8 %, 25.4 %, and 44.8 % respectively. It is evident that the removal efficiency of HA reactor for component 1 is comparatively lower, and the aromatic-like protein II is recalcitrant organic compounds. Furthermore, MAHA-2 demonstrates a substantial advantage in the removal of component 1.

3.3. Assessment of microbial metabolism and activity

3.3.1. Live/dead percentage

Live/dead cells of the sludge in the reactor were observed through staining with live/dead cells staining agents under laser scanning confocal microscope. Major dead cells are presented in HA and MAHA-1 samples, whereas more live cells can be observed in the MAHA-2 image (Fig. 6(a-c)). Subsequently, the fluorescence ratio of live/dead cells was analyzed by ImageJ, and the live cells proportions of sludge in HA, MAHA-1 and MAHA-2 reactors are 28.96 %, 31.89 % and 49.32 %, respectively. This demonstrated that MAHA-2 mitigated wastewater toxicity due to the judiciously calibrated DO concentration, resulting in the enhanced number of live cells in the sludge and augmenting microbial activity. The microbial cell morphology in the reactor was further visualized by SEM. Notably, higher biotoxicity could cause oxidative damage to cytomembrane, resulting in microbial cell rupture (Zheng et al., 2018). It shows that cells in all reactors suffer varying degrees of rupture, with MAHA-2 exhibiting the least damage and thereby signifying comparatively reduced wastewater toxicity to cells (Fig. 6(d-f)).

3.3.2. Functional enzyme activity

Microbes in UASB reactor typically required large amounts of enzymes when confronted with the degradation of refractory organics (e.g., phenols). Phenol hydroxylases and catechol 1,2-dioxygenases, intricately associated with phenolic degradation, emerged as pivotal enzymatic catalysts in the biodegradation (Wang et al., 2024). Under the presence of O_2, phenol was converted to catechol by the catalysis of phenol hydroxylase. And this transformation was followed by successive ring-opening and chain cleavage processes, facilitated by the engagement of 1,2-dioxygenase. Ultimately, the HA process and degradation of phenolic constituents were achieved through the concerted involvement of diverse dehydrogenases. Throughout the organic degradation

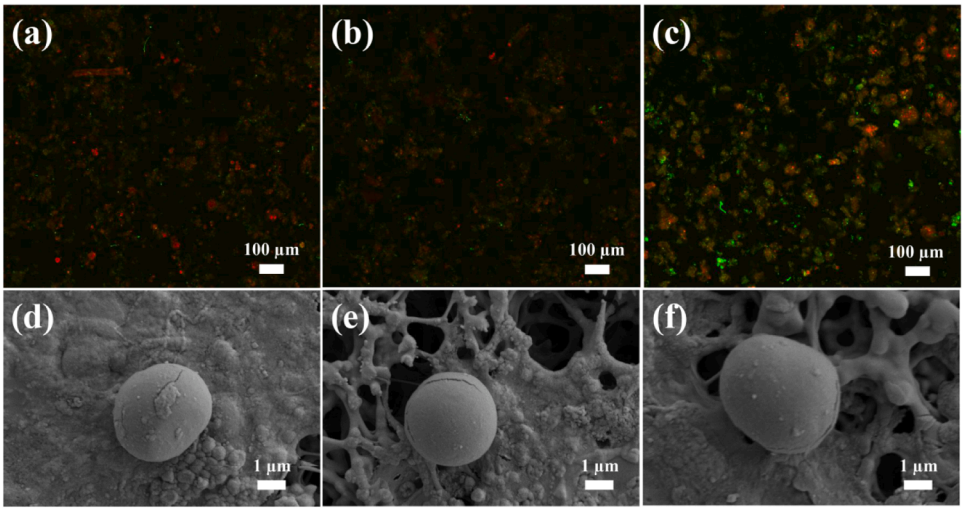


Fig. 6. Images of the sludge dead and live cell staining of (a) HA, (b) MAHA-1 and (c) MAHA-2 samples; SEM images of microbial cell of (d) HA, (e) MAHA-1 and (f) MAHA-2 samples.

process, ATP provided the energy needed for microbial growth and metabolic activity (Wang and Zhou, 2025). Our focus converged upon the concentrations of phenol hydroxylase, catechol 1,2-dioxygenase, dehydrogenase, and ATP under varying DO conditions (Fig. S10). Remarkably, the concentrations of these enzymes in HA and MAHA-1 reactors exhibit comparability. This implied that lower DO concentration failed to exert a substantial promotive influence upon the functional enzyme activity of microbial system. However, the presence of OUR in the aerobic facultative bacteria resulted in a slight increase in ATP production. Conversely, MAHA-2 displayed a remarkable increase in the concentrations of all functional enzymes and ATP, demonstrating that the microbes produced more ATP and functional enzyme, thus improving organics degradation (Zhang et al., 2022).

3.3.3. *Variation of EPS*

The predominant component of EPS is protein in all reactors (Fig. 7 (a, b)). Notably, there is a marked escalation in the EPS concentration of Stage III (the sum of polysaccharides, proteins, and humic acids) compared to Stage I. This may be attributed to active secretion or passive release by microbes, where active secretion refers to microbes actively synthesizing EPS to form a physical barrier or chelating toxic substances by up-regulating the relevant genes under the pressure of toxic substances or C/N imbalance (Tao et al., 2025), and passive release refers to the release of intracellular substances due to damage to the cell membrane when toxicity exceeds the cellular tolerance threshold, or the lysis of EPS along with intracellular components after cellular autolysis (Cao et al., 2021). EPS consists of soluble EPS (S-EPS), LB-EPS and TB-EPS, of which S-EPS exists in the liquid phase and originates from substrate metabolism and the release of intracellular substances after cellular senescence and death (Yang et al., 2022), while LB-EPS and TB-EPS are secreted by microbes, partly from cellular adsorption and organic transformation (Wu et al., 2020). Meanwhile, the contents of LB-EPS and TB-EPS including polysaccharides, proteins and humic acids therein are significantly higher in MAHA-2 at Stage III than in other reactors. Combined with the results of live/dead cells, MAHA-2 has a significantly higher proportion of live cells at Stage III than other reactors. It is confirmed that although increased wastewater concentration and toxicity accelerates cell death to release part of the EPS, and that the EPS may be derived from cellular adsorption and organic transformation, the microbial stress secretion mechanism of "active barrier" is dominant at Stage III in MAHA-2.

It is noteworthy that TB-EPS content significantly surpasses that of LB-EPS. This discrepancy arose from the fact that LB-EPS was porous, displaying a greater correlation with suspended or colloidal organics in wastewater (Wei et al., 2024). Moreover, LB-EPS was more susceptible to environmental fluctuations, rendering it unstable. Remarkably, the composition and concentration of LB-EPS and TB-EPS in MAHA-1 aligns closely with that in HA. However, with a further elevation in DO concentration, there is a conspicuous augmentation in LB-EPS and TB-EPS concentrations, particularly in term of protein concentration.

In comparison to anaerobic condition, the presence of O_2 fostered the pace of microbial growth and metabolism, resulting in the increased secretion of LB-EPS and TB-EPS. This fortified their resilience against wastewater toxicity and enhanced their capacity for pollutant sequestration. Notably, proteins played a pivotal role in the microbial capture and metabolic processing of contaminants. Phenolic compounds carried a negative charge due to weak ionization, rendering them prone to electrostatic interaction with positively charged amino functional

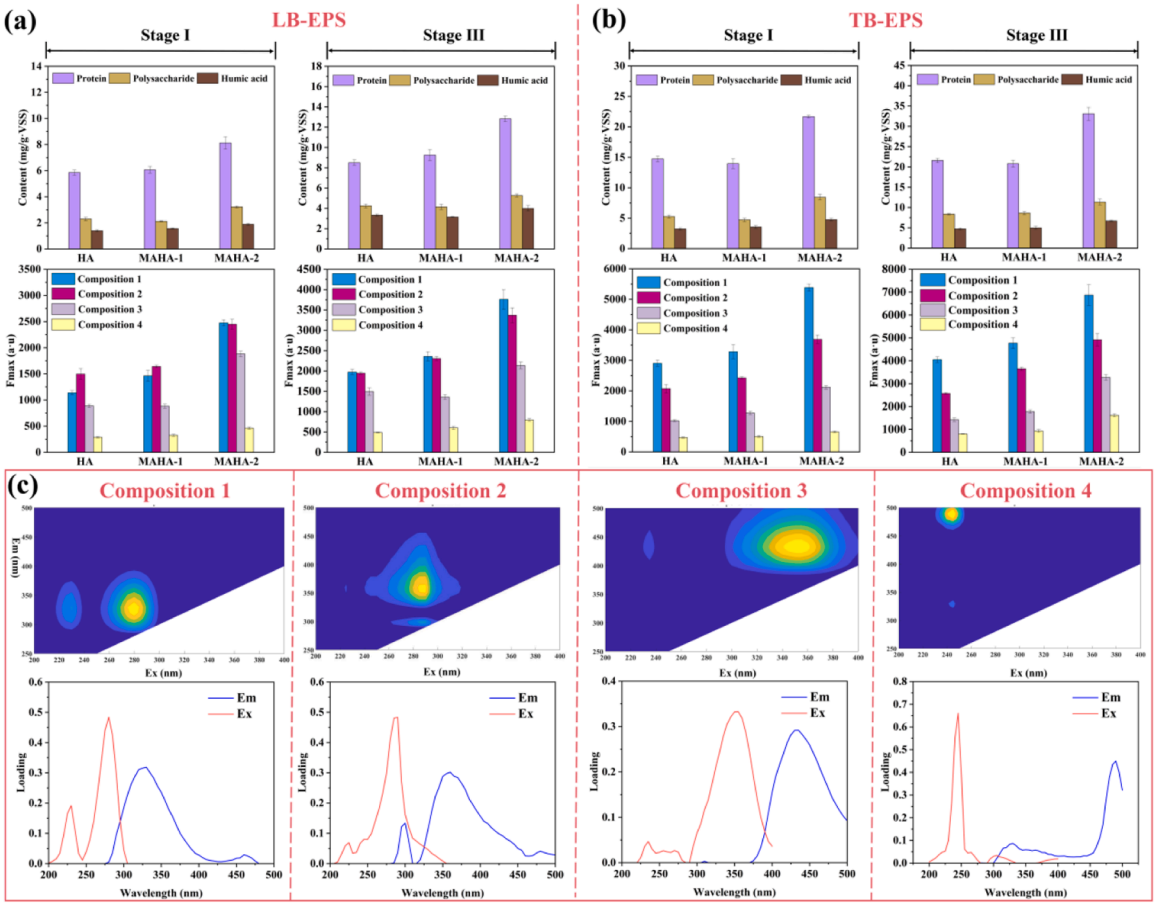


Fig. 7. Variations of (a) LB-EPS content and (b) TB-EPS content in Stage I and Stage III; (c) 3D-EEM spectra of four fluorescence components with PARAFAC analysis at various DO concentrations.

groups in proteins (Li et al., 2021). Consequently, these compounds were captured by cells and underwent subsequent metabolic processes. Simultaneously, hydrophobic organic substances (e.g., heterocyclic compounds and long-chain alkane) readily formed associations with hydrophobic proteins through hydrophilic-hydrophobic interactions (Park et al., 2025; Xie et al., 2020). Furthermore, an array of functional enzymes constituted an integral component of proteins. The augmentation in protein content implied a simultaneous increase in the levels of functional enzymes associated with pollutant degradation.

There are four main fluorescent components of LB-EPS and TB-EPS in different reactors according to PARAFAC analysis. As shown in Fig. 7(c), component 1 (Ex/Em 230 or 280 nm/330 nm) represents tyrosine-like aromatic proteins and microbial byproduct-like substances; Component 2 (Ex/Em 290/300 or 370 nm) indicates microbial metabolites; Component 3 (Ex/Em 355/435 nm) denotes humic acid-like; Component 4 (Ex/Em 245/490 nm) is ascribed to fulvic acid-like (Wang et al., 2022). F_{max} of four components of EPS in reactor were further analyzed by PARAFAC. Among them, the F_{max} of component 1 is the largest, followed by component 2, and components 3 and 4 are lower. It shows the proportion of protein-like and microbial metabolite-like fluorescent substances in the LB-EPS and TB-EPS is relatively high, with the highest content of protein-like fluorescent fractions. And this result is consistent with the EPS composition analysis.

As the influent COD load is increased from Stage I to Stage III, the F_{max} of each fluorescence component in LB-EPS and TB-EPS also improves, which is consistent with the increase in EPS concentration at high load. Then, the F_{max} (DO: 0–0.1 mg/L) of each fluorescent component is slightly higher than that of HA reactor (Fig. 7(a, b)). By contrast, F_{max} of each fluorescent component is highest in the MAHA-2 reactor for TB-EPS and LB-EPS, especially for the protein-like F_{max} of component 1. Because higher DO concentration promoted the content of protein in EPS, and protein was related to the capture of pollutants and the content of functional enzymes by microbes, thus promoting the CGW treatment capacity (Hu et al., 2019).

3.4. Microbial community analysis

The microbial community of reactors was revealed using the Illumina MiSeq high-throughput sequencing method. Based on the Shannon dilution curve at the genus level, it can be observed that the curves of all sample gradually flatten (Fig. S11), indicating that the sequencing data sufficiently represent the entire community. Table S5 presents the community abundance and alpha diversity indices for each sample. The coverage indices are all above 0.995, indicating that the sequence analysis adequately represents the complete microbial community (Uludag-Demirer et al., 2025). The Ace and Chao indices serve to evaluate the total species count, indicating species richness, with higher value clearly signifying an increase in species richness (Jena et al., 2016). The Shannon index integrates species richness and evenness, with higher value denoting an increase in diversity (Wu et al., 2023; Zou et al., 2020). Besides, the Simpson reflects the community dominance, applied as a measure of community diversity, which represents a combination of species richness and evenness in a given ecosystem. When few dominant species are extremely prominent in the system, Simpson value increases and diversity decreases. Conversely, when the distribution of species is balanced, Simpson value declines and diversity rises (Jiang et al., 2025; Qin et al., 2024). The Shannon, ACE, and Chao indices of MAHA-2 are higher than those of HA and MAHA-1 reactors. It shows that the richness and diversity of microbes in the reactor are higher at DO of 0.2–0.3 mg/L. The Rank-Abundance curve also supports this conclusion (Fig. S12). Additionally, the Simpson index of MAHA-2 is lower than that of MAHA-1 and HA reactors, indicating the increase in microbial community diversity due to the elevated DO. This not only enhanced microbial ecological stability but also strengthened the ability to withstand high-toxicity environmental pressures from CGW. It is also one of the reasons for the superior pollutant removal capability of MAHA-2.

In the microbial community at the phylum level, *Proteobacteria, Firmicutes, Actinobacteriota, Bacteroidota, Desulfobacterota, Chloroflexi* and *Synergistota* emerge as the dominant phyla across all reactors (Fig. 8 (a)). The key distinction between microbial phyla in anaerobic and micro-aeration reactors lies in their relative abundances. The relative abundances of microbial phyla in MAHA-1 and HA reactors exhibit a resemblance, signifying that lower DO concentration has limited impact on the microbial ecology of anaerobic system. However, as DO concentration increases, significant shifts occur in the relative abundances of dominant phyla. The relative abundances of *Proteobacteria, Firmicutes, Actinobacteriota, Bacteroidota, Desulfobacterota, Chloroflexi*, and *Synergistota* are 53.32 %, 17.48 %, 5.69 %, 3.88 %, 6.24 %, 0.72 % and 1.58 %, respectively, in HA reactor. In contrast, MAHA-2 exhibits relative abundances of 45.62 %, 20.70 %, 7.36 %, 5.65 %, 1.45 %, 2.80 % and 2.93 % for the same phyla. Notably, *Firmicutes, Actinobacteriota, Bacteroidota, Chloroflexi*, and *Synergistota* exhibit higher relative abundances in MAHA-2. Among them, *Firmicutes, Bacteroidetes*, and *Chloroflexi* were crucial phyla involved in the hydrolysis of organics by producing extracellular hydrolases to hydrolyze complex organics to VFAs (Bovio et al., 2019; Song et al., 2019).

Proteobacteria, predominantly facultative or obligate anaerobes in wastewater treatment systems, mostly consist of heterotrophic bacteria that play a crucial role in COD removal, denitrification and desulfurization processes (Liu et al., 2022). However, the relative abundances of *Proteobacteria* and *Desulfobacterota* in MAHA-2 have decreased by 7.71 % and 4.79 %, respectively. The micro-oxygen environment might inhibit the growth of obligate anaerobes, leading to a decline in the relative abundance of *Proteobacteria. Desulfobacterota*, associated with sulfur cycling, primarily encompassed sulfate-reducing bacteria (Song et al., 2021). The micro-oxygen condition could inhibit the growth of sulfate-reducing bacteria, resulting in a decrease in the relative abundance (Zhang et al., 2021). Therefore, at the phylum level, higher DO concentration promoted the enrichment of phyla associated with the HA of organics, thereby facilitating the biodegradation of organics in wastewater.

The differences in the functional microbial community at the genus level under different DO concentrations were further explored. *Ottowia* (7.96 %), *Trichococcus* (6.33 %), *Soehngenia* (3.69 %), *Lentimicrobium* (4.76 %) and *Desulfobulbus* (5.44 %) are the genera with the highest relative abundances in the HA reactor (Fig. 8(b)). *Ottowia*, one of facultative anaerobes, could participate directly in denitrification using aromatic compounds as electron donors (Huang et al., 2024). Organic pollutants in CGW are mainly dominated by phenols, thus *Ottowia* has the highest relative abundance and plays a key role in organics degradation. *Trichococcus* and *Soehngenia* are also facultative anaerobes, belonging to the phylum *Firmicutes*, which are primarily associated with the organics hydrolysis for acid production (Grouzdev Denis et al., 2019; Ji et al., 2020; Zuo et al., 2023). For instance, they could hydrolyze and ferment glucose, sucrose and lactose to produce various VFAs (e.g., lactic acid, acetate, formate, and other acids).

Lentimicrobium and *Desulfobulbus* are strict anaerobes belonging to the phyla *Bacteroidota* and *Desulfobacterota*, respectively (Sun et al., 2016; Suzuki et al., 2007). *Lentimicrobium* is associated with anaerobic fermentation of organics (Zhen et al., 2023), while *Desulfobulbus* is a typical sulfate-reducing bacterium that reduces sulfate in wastewater to produce toxic H_2S (Tian et al., 2017). Compared with HA reactor, there is little change in the relative abundance of the aforementioned five major microbial genera in MAHA-1 reactor. However, *Ottowia, Trichococcus*, and *Soehngenia* have increased relative abundances to 11.69 %, 7.94 %, and 5.50 % respectively in MAHA-2. *Lentimicrobium* and *Desulfobulbus* are inhibited by higher DO concentration, with a consequent decrease in relative abundance from 4.76 % and 5.44 % to 1.20 % and 0.67 %, respectively. Micro-oxygen environment significantly suppressed the growth of sulfate-reducing bacteria (e.g., *Desulfobulbus*) and were commonly used to control the production of H_2S during the

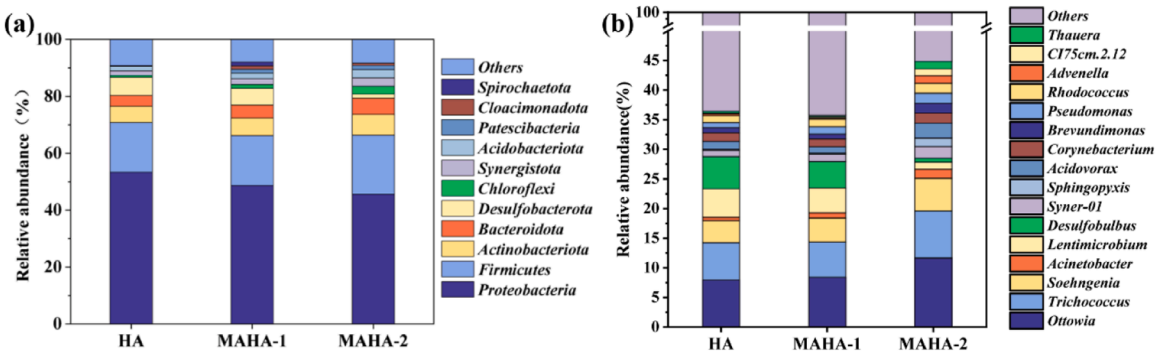


Fig. 8. Relative abundance of the dominant bacterial community at (a) phylum level and (b) genus level.

anaerobic digestion process (Mahdy et al., 2020). Additionally, in MAHA-2 reactor, there is a certain degree of increase in the relative abundances of facultative bacteria associated with fermentation and genera related to the degradation of phenols, polycyclic aromatic hydrocarbons, and long-chain alkanes compared to HA reactor. For example, the relative abundances of *Syner-01, Corynebacterium, Brevundimonas* and *CI75cm.2.12*, associated with fermentation and acid production, have increased from 1.03 %, 1.42 %, 0.88 %, and 0.12 % to 1.99 %, 1.69 %, 1.65 %, and 1.18 %, respectively. *Sphingopyxis, Acidovorax, Pseudomonas* and *Advenella*, associated with the anaerobic hydrolysis of long-chain alkanes and benzenes (Dou et al., 2024; Monferrán et al., 2005; Verma et al., 2015; Wang et al., 2016), have increased their abundances from 0.19 %, 1.35 %, 0.92 %, and 0.27 % to 1.40 %, 2.52 %, 1.73 %, and 1.27 %, respectively. As a result that the hydrolysis rate of long-chain alkanes under micro-oxygen condition is faster. For example, *Pseudomonas* degraded *n*-alkanes (C11-C21) at a faster rate and produced more acetate (Rocha et al., 2011). Furthermore, genera such as *Acinetobacter, Rhodococcus* and *Thauera*, associated with the degradation of phenols or polycyclic aromatic hydrocarbons (Ma et al., 2023; Mao et al., 2010; Szilveszter et al., 2023), have also shown increased abundance. In conclusion, the reactor with higher DO concentration has enriched genera associated with the fermentation for acid production, as well as the degradation of phenols, polycyclic aromatic hydrocarbons, and long-chain alkanes. And this enrichment facilitated the HA and ultimate degradation of recalcitrant organics. The evidence highlights the crucial role of micro-oxygen environment in enhancing the organics removal in actual CGW treatment.

3.5. Macrogenomics analysis

3.5.1. The links between microbial metagenomes and aromatic metabolic profile

Aromatic compounds have conjugated ring structure and the delocalization property of π orbitals, rendering the benzene rings exceptionally stable and resistant to microbial ring-opening cleavage (Nzila, 2018). Thus, the ring-opening cleavage of benzene structures represents a pivotal rate-limiting step in the degradation of aromatic organic compounds by anaerobes (Boll et al., 2002). Simultaneously, it is a critical facet for reducing the toxicity of aromatic organics and enhancing wastewater biodegradability (Kuroda et al., 2022; Zhao et al., 2018). Under the strict anaerobic condition, the system lacked ring-opening activators (ring-opening enzymes substrates) and terminal electron acceptors (e.g., O_2, NO_3^--N, sulfate, Fe^{3+}, and high-valence Mn) (Chen et al., 2023; van Leeuwen et al., 2022; Zhao et al., 2025). This circumstance forced microbes to engage in ring-opening through slower and lower-capacity reduction pathways, such as the methane reduction pathway. Conversely, O_2 acted as the activator of benzene ring-opening reaction through the action of various monooxygenases or dioxygenases in the micro-oxygen environment (Zhang et al., 2021). This process weakened the stability of C—C bonds in the benzene rings, making them more susceptible to cleavage and gradual transformation into small molecules that were readily utilized by microbes (Zhang et al., 2022). In both anaerobic and micro-oxygen conditions, a variety of ring-opening functional enzymes and related microbes played pivotal roles in the auto-catalytic process (Zhu et al., 2023).

The Kyoto Encyclopedia of Genes and Genomes (KEGG) database was utilized to screen and analyze the metagenome of microbial community, and the links between dominant microbial species (>0.5 %) and ring-opening cleavage enzymes were constructed (Fig. S13). As depicted in Fig. 9(a), the ring-cleavage enzyme was listed with related dominant species. Under the presence of DO, aromatic organic compounds in CGW primarily underwent eight distinct ring-opening degradation pathways. The pivotal ring-opening cleavage enzymes involved catechol 1,2-dioxygenase, catechol 2,3-dioxygenase, gentisate 1,2-dioxygenase, protocatechuate 3,4-dioxygenase, homogentisate 1,2-dioxygenase, 3,4-dihydroxyphenylacetate 2,3-dioxygenase, 2,3-dihydroxyphenylpropionate 1,2-dioxygenase and biphenyl-2,3-diol 1,2-dioxygenase. A total of 15 microbial species were found to be associated with the expression of the above gene-encoding ring-opening cleavage enzymes (Boll et al., 2002; Nzila, 2018). Among them, the catechol 1,2-dioxygenase, catechol 2,3-dioxygenase, protocatechuate 3,4-dioxygenase, and gentisate 1,2-dioxygenase possessed relatively abundant species, suggesting that they played the central role in the micro-oxygen ring-opening cleavage of aromatic organic compounds. Conversely, under anaerobic condition, aromatic organic compounds underwent ring-opening cleavage through 2-ketocyclohexanecarboxyl-CoA hydrolase, benzoyl-CoA-dihydrodiol lyase, and 1,4-dihydroxy-2-naphthoyl-CoA hydrolase. It's evident that the number of microbial species involved in encoding is significantly lower than micro-oxygen condition. Notably, the microbial species are closely related to 2-ketocyclohexanecarboxyl-CoA hydrolase, indicating a greater involvement of this enzyme in the anaerobic ring-opening cleavage pathway. Fig. S14 shows the detailed metabolic pathways and specific information regarding ring-opening enzymes. Therefore, under micro-oxygen conditions, both the number of microbial species associated with ring-opening enzymes and ring-opening metabolic pathways are far greater than anaerobic condition. Additionally, the sources of ring-opening enzymes are more diverse, thus promoting the ring-opening and metabolism of aromatic organic compounds.

Subsequently, we further conducted the quantitative analysis about the contributions of key ring-opening enzymes at the genetic and microbial species levels for the degradation pathways of aromatic organic compounds. The abundance of key functional enzyme genes involved in the ring-opening processes was assessed at the KEGG orthology by the Reads Per Kilobase of exon model per Million mapped reads (RPKM) (Fig. 9(b)). The genes abundance of ring-opening enzymes associated from HA to MAHA-2 is noticeably higher. DO, serving as an electron acceptor, not only expedited organics degradation but also generated more energy to support the metabolism and proliferation of facultative microbial species associated with the expression of micro-oxygen ring-opening enzymes (Ghosh and Mukherji, 2021; Zhou et al., 2024). This

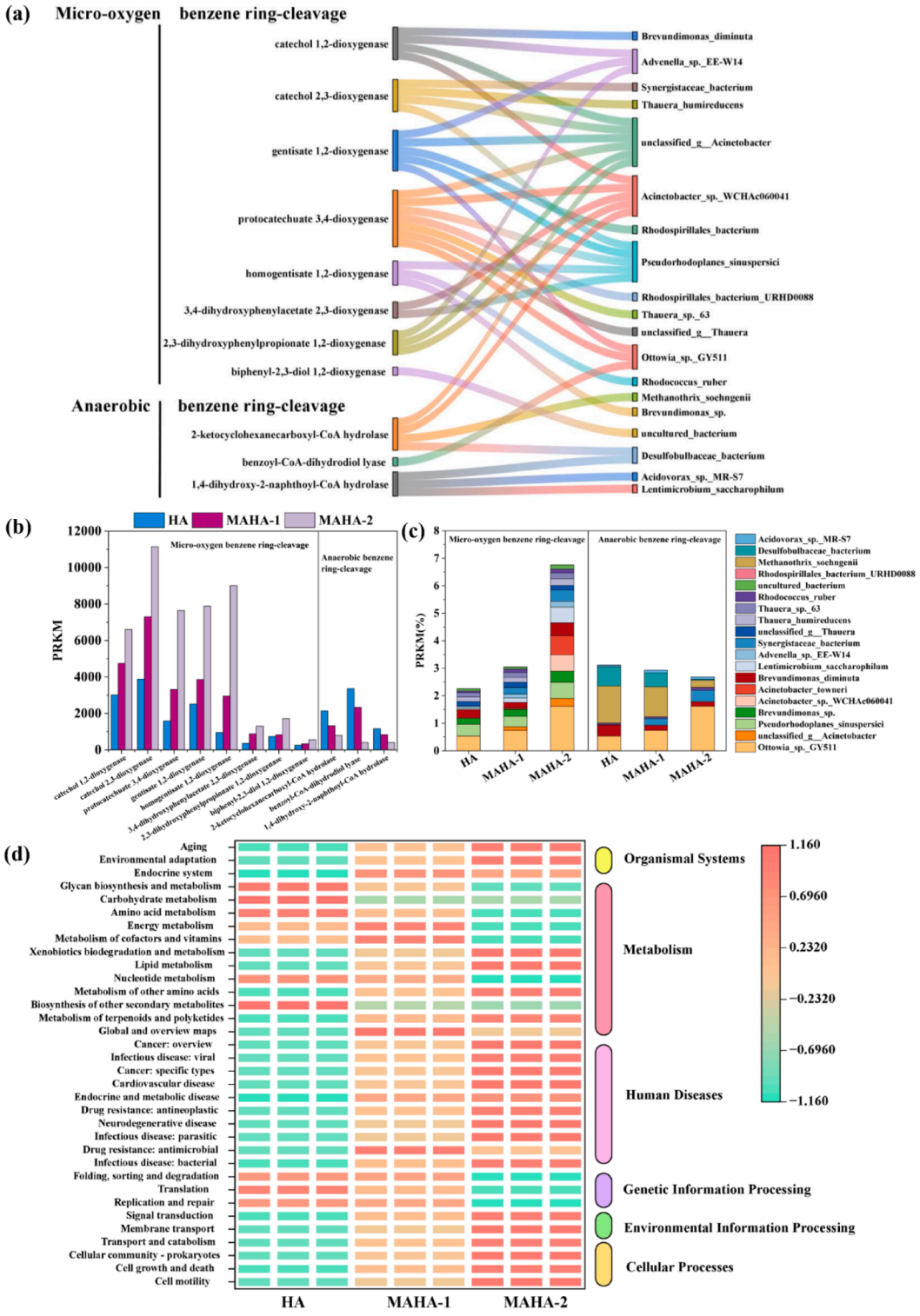


Fig. 9. (a) The distribution of ring-cleavage enzyme in related dominated species; (b) The abundance of key ring-opening cleavage enzymes, and (c) the abundance of encoding dominated ring-opening species in micro-oxygen and anaerobic conditions; (d) Expression of key KEGG pathways at level 1 and level 2 in HA, MAHA-1 and MAHA-2 reactors.

resulted in a higher abundance of microbial species related to the encode of micro-oxygen ring-opening enzymes in MAHA-2 than in HA reactor by 4.49 %, thereby producing more micro-oxygen ring-opening enzyme genes. In addition to the elevated abundance of facultative microbes that contributed to both anaerobic and micro-oxygen ring-opening enzymes (e.g., *Ottowia*_sp._*GY511*), other microbes that were only associated with the encode of anaerobic ring-opening enzymes (e.g., *Methanothrix_soehngenii* and *Desulfobulbaceae_bacterium*) underwent oxidative inhibition in the presence of elevated DO concentrations. This resulted in a slight decrease in the overall abundance of microbial species associated with the encode of anaerobic ring-opening enzymes.

The cumulative abundance of various micro-oxygen ring-opening enzyme genes in MAHA-2 (405,868) significantly surpasses that in HA reactor (6670). Correspondingly, the total abundance of related

microbial species is also higher in MAHA-2 (6.75 %) compared to HA reactor (3.11 %) (Fig. 9(c)). This suggested that the micro-aeration reactor existed a greater potential for microbial expression of ring-opening enzymes, consequently resulting in a more robust capacity for the ring-opening and degradation of aromatic organic compounds. In MAHA-2, the cumulative RPKM values for micro-oxygen and anaerobic ring-opening enzyme genes are 405,868 and 1630, respectively. It is evident that microbial degradation of aromatic organic compounds is primarily through the micro-oxygen pathway in MAHA-2. Among these enzymes, the genes for catechol 1,2-dioxygenase, catechol 2,3-dioxygenase, protocatechuate 3,4-dioxygenase, gentisate 1,2-dioxygenase, and homogentisate 1,2-dioxygenase exhibit the highest abundance and together account for 99.1 % of the total micro-oxygen ring-opening enzyme genes, which are mainly involved in the ring-degradation pathway of aromatic organics in MAHA-2. In summary, MAHA-2 reactor had the highest ring-opening enzyme species associated with aromatic organics degradation. This caused a higher potential for the production of multiple ring-opening enzymes, resulting in greater catabolism and degradation of aromatic organics.

3.5.2. KEGG pathway annotation of upregulated proteins

The open reading frames annotated in the KEGG database were utilized to compare the relative abundance of potential functional genes among the reactors. The KEGG pathway enrichment analysis reveals that the abundance of 34 distinct functional categories at level 2 exceeds 0.1 %. In comparison to HA, more than half of these functional categories exhibit a slight increase in abundance in MAHA-1, with MAHA-2 demonstrating the highest levels of abundance (Fig. 9(d)). These functional categories are associated with amino acid metabolism, carbohydrate metabolism, and energy metabolism. Amino acids are pivotal cellular constituents, participating in growth and various metabolic processes (Song et al., 2025). In MAHA-2, the abundances of amino acid and carbohydrate-related metabolic functions are 8.84 % and 9.24 %, respectively, surpassing MAHA-1 (7.86 %, 8.40 %) and HA (7.00 %, 8.31 %). This indicates the activation of species related to amino acids and carbohydrates, thereby enhancing the removal efficiency of organic pollutants (e.g., phenols, aromatic compounds, etc.). As depicted in Fig. S15, MAHA-2 exhibits relatively higher abundances of genes associated with carbon metabolism and amino acid biosynthesis compared to MAHA-1 and HA reactors. The above results underscore a significant increase in the abundance of functions related to microbial growth and the metabolism of organic compounds in MAHA-2, thereby conferring a heightened capacity for the organics removal.

3.5.3. Analysis of hydrolase genes related to organics hydrolysis

The hydrolysis process holds significant importance in reducing wastewater toxicity and enhancing biodegradability. In this process, organic acids were formed when water molecules came into contact with organic compounds catalyzed by the relevant hydrolases, resulting in the cleavage of chemical bonds (e.g., peptide, ester, and anhydride bonds) in the organic molecules (Song et al., 2019). Thus, the analysis of hydrolytic genes contributed to understanding the hydrolysis process. VFAs molecules bound to water were catalyzed by ester bond hydrolases. The cleavage of ester bond released VFAs that promoted the growth and metabolism of other microbes. The abundance of ester bond hydrolases genes is depicted in Fig. S16. The relative abundances of ester bond hydrolases in all reactors are as follows: [EC: 3.1.21.3] hsdS (15, 924, 22,900, 28,916), hsdR (21,892, 35,688, 40,748), and [EC: 3.1.-.-] recJ (15,812, 22,206, 23,416). It is evident that these ester bond hydrolases genes exhibit higher abundance in MAHA-2 compared to MAHA-1 and HA reactors. The hydrolysis rate of VFAs molecules was positively correlated with the abundance of ester bond hydrolases, indicating that MAHA-2 possessed a higher abundance of these genes encoding ester bond hydrolases, resulting in a stronger ability to hydrolyze organic compounds. Furthermore, the hydrolysis of peptide bonds was a vital biochemical reaction. Peptide bonds, under the catalysis of peptidases, underwent hydrolysis with the generation of carboxylic acids and amino groups (Gao et al., 2023; Guo et al., 2021). There are multiple types of peptidases, with the abundances of various peptidase genes as follows: [EC: 3.4.19.13] ggt (14,514, 22,232, 34, 476), [EC: 3.4.-.-] K08303 (10,926, 19,532, 21,304), [EC: 3.4.24.-] ftsH, hflB (21,408, 23,686, 22,818), pqqL (24,708, 23,164, 20,172), [EC: 3.4.21.53] lon (24,116, 27,178, 27,538), [EC: 3.4.16.4] mrcA (16,492, 24,584, 37,930), and [EC: 3.4.11.9] pepP (17,858, 18,552, 20,160). The abundances of these peptide bond hydrolases genes are significantly higher in MAHA-2 than in MAHA-1 and HA reactors, indicating that the microbial community in MAHA-2 possesses a stronger capability to hydrolyze peptide bonds, which promotes microbial growth and increases microbial community abundance. Finally, anhydride hydrolases catalyzed the hydrolysis of anhydride bonds in organic compounds, yielding the corresponding carboxylic acids and alcohols. The abundances of anhydride hydrolases in all reactors are as follows: [EC: 3.6.3.-] moxR (16,478, 22,864, 23,778), [EC: 3.6.4.-] priA (18,650, 18, 700, 19,376), [EC: 3.6.4.12] uvrD, pcrA (34,792, 35,094, 33,500), and recG (24,468, 28,522, 28,344). These anhydride hydrolases exhibited variations in different reactors, because the anaerobic, micro-oxygen, and facultative bacteria could participate in anhydride bond hydrolysis. Similarly, anhydride hydrolases indirectly enhanced biological activity. In summary, the overall abundance of peptide, ester, and anhydride bond hydrolases genes in MAHA-2 surpassed that of MAHA-1 and HA reactors. This suggested that micro-oxygen environment contributed to the overall ability of the microbial community to hydrolyze organic compounds and produce organic acids, and indirectly enhanced the overall biological activity of community.

3.5.4. Biodegradation pathway of phenol

Although qualitative and quantitative relationships between encoding key ring-opening enzymes for aromatic organic compounds and functional species have been established through metagenomic analysis, a deeper understanding of various functional enzymes in their role and metabolic pathways for participating in the biological degradation of organic substances is needed. Phenolic substances constituted 60–80 % of total organic pollutants in CGW, with phenol being the highest proportion among phenolic organics. Hence, we selected phenol as the representative toxic and refractory aromatic organic, and the KEGG database was applied for exploring the metabolic pathways of phenol and the involved key functional enzymes. As shown in Fig. 10(a), the biological degradation pathways of phenol can be divided into micro-oxygen pathway A and anaerobic pathway B. In pathway A, the transformation of phenol to catechol is one of the rate-limiting steps during phenol degradation (Sabri et al., 2025). Under the action of [EC: 1.14.13.243;1.14.13.244] phenol/toluene 2-monooxygenase, O atoms participated in the activation reaction of phenol to produce catechol via hydroxylation (Zhang et al., 2023a). The genes encoding [EC: 1.14.13.243;1.14.13.244] phenol/toluene 2-monooxygenase in MAHA-2 consist of five parts, with relative abundances of 2310, 472, 2396, 740 and 2304, respectively, significantly higher than in MAHA-1 (1426, 248, 1480, 566, 1630) and HA (970, 238, 1110, 470, 1224) reactors (Fig. 10(b)). This indicated that the micro-oxygen environment in MAHA-2 promoted an increase in the gene abundance of phenol/toluene 2-monooxygenase, facilitating the activation of phenol and subsequent degradation reaction.

The micro-oxygen degradation pathway of catechol was divided into pathway A-1 and A-2. In pathway A-1, catechol underwent oxygen molecular insertion under the action of [EC: 1.13.11.1] catechol 1,2-dioxygenase, resulting in the ring-opening cleavage of catechol to generate *cis*,cis-muconic acid. Subsequently, in combination with [EC: 5.5.1.1] muconate cycloisomerase, muconic acid underwent intramolecular cleavage to generate muconolactone (Skiba et al., 2002). Subsequently, under the action of [EC: 5.3.3.4] muconolactone D-isomerase, intramolecular redox led to the generation of 3-oxoadipate-enol-lactone (Li et al., 2023). Finally, with the successive action of

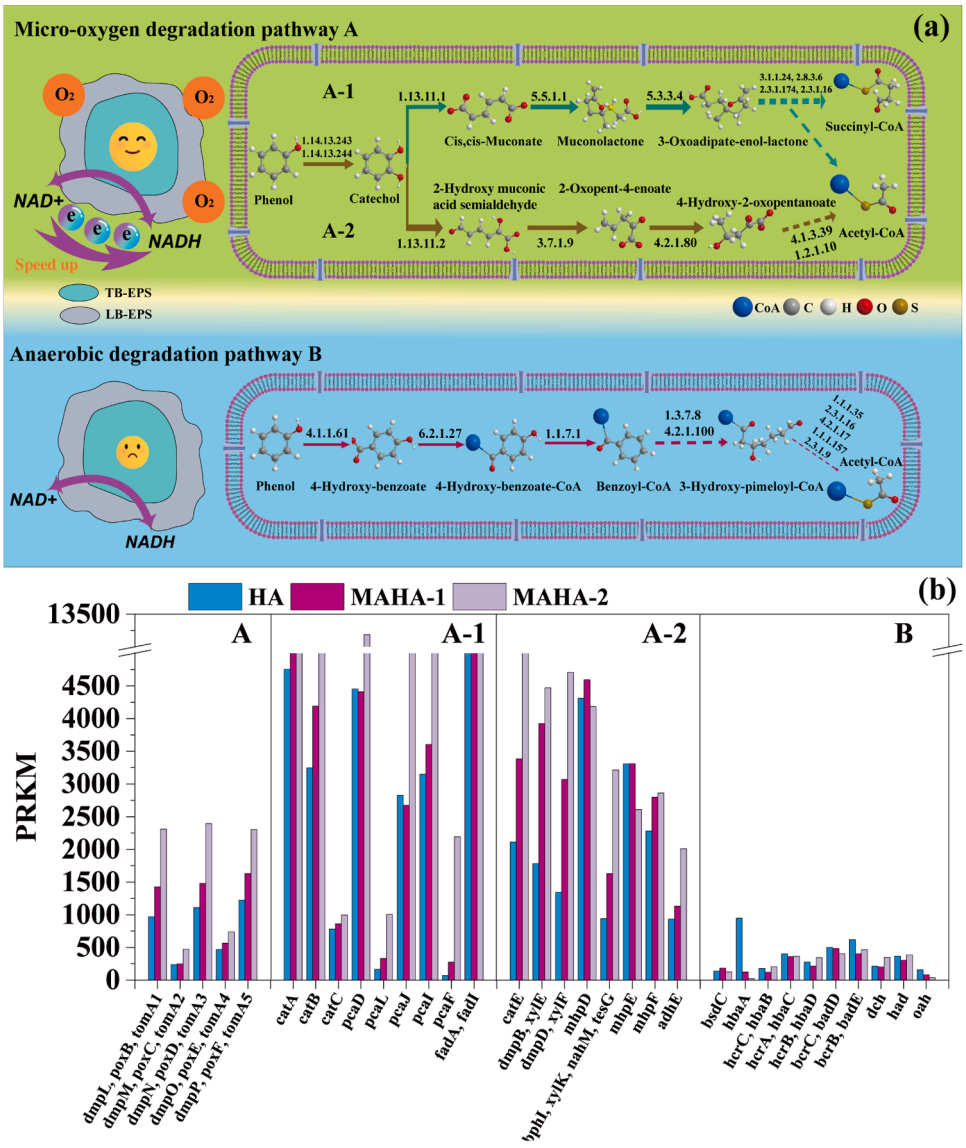


Fig. 10. (a) Metabolic pathways of phenol by KEGG analysis; (b) the abundance of functional enzyme genes associated with phenol metabolic pathways.

[EC: 3.1.1.24] 3-oxoadipate enol-lactonase, [EC: 2.8.3.6] 3-oxoadipate CoA-transferase, [EC: 2.3.1.174] 3-oxoadipyl-CoA thiolase and [EC: 2.3.1.6] acetyl-CoA acyltransferase, succinyl-CoA and acetyl-CoA were progressively decomposed and entered the tricarboxylic acid cycle, and mineralized into CO_2 and H_2O (Mu et al., 2022). In pathway A-2, catechol underwent ring-opening cleavage through oxygen molecular insertion with the action of [EC: 1.13.11.2] catechol 2,3-dioxygenase, forming 2-hydroxymuconic semialdehyde. Subsequently, after the action of [EC: 3.7.1.9] 2-hydroxymuconate-semialdehyde hydrolase, C—C bonds were broken to form 4-oxalocrotonate. Combining with [EC: 4.2.1.80] 2-keto-4-pentenoate hydratase, the C—O bonds were broken to generate 4-hydroxy-2-oxopentanoate. Lastly, acetaldehyde was produced by the action of [EC: 1.2.1.10] 4-hydroxy 2-oxovalerate aldolase, and NAD^+ or $NADP^+$ were used as the electron acceptors to further generate acetyl-CoA and enter the tricarboxylic acid cycle (Varjani, 2017). Pathway B was the anaerobic degradation pathway of phenol, wherein phenol underwent the action of [EC: 4.1.1.61] vanillate/4-hydroxybenzoate decarboxylase to generate 4-hydroxybenzoic acid, which then combined with [EC: 6.2.1.27] 4-hydroxybenzoate-CoA ligase to generate 4-hydroxybenzoyl-CoA. [EC: 1.1.7.1] 4-hydroxybenzoyl-CoA reductase further converted 4-hydroxybenzoyl--CoA to benzoyl-CoA. Subsequently, with a series of enzymatic actions including [EC: 1.3.7.8] benzoyl-CoA reductase, [4.2.1.100] cyclohexa-1,5-dienecarbonyl-CoA hydratase, [EC: 1.1.1.368] 6-hydroxycyclohex-1-ene-1-carbonyl-CoA dehydrogenase, and [EC: 3.7.1.21] 6-oxocyclohex-1-ene-carbonyl-CoA hydrolase, acetyl-CoA was generated to enter the tricarboxylic acid cycle and decomposed into CO_2 and H_2O (Owusu-Agyeman et al., 2021).

The expression of functional enzymes was directly regulated and encoded by genes, thus the abundance of genes encoding key functional enzymes at each step in the metabolic pathway played a key role in the biodegradation of phenol (Table S6). The gene abundance associated with key functional enzymes in the metabolism pathways of A-1, A-2 and B can be observed in Fig. 10(a). In the anaerobic degradation pathway of phenol, the gene abundance related to functional enzymes is significantly lower than that in the micro-oxygen pathway. This indicates that there is a greater potential for functional enzymes related to phenol degradation in the presence of O_2, leading to the enhanced phenol degradation capability. Meanwhile, in the pathways A-1 and A-2, the gene abundance of functional genes is slightly elevated in MAHA-1 compared to HA reactor, and notably increased in MAHA-2 reactor. For instance, the gene abundance of key functional genes involved in pathway A-1, such as [EC: 1.13.11.1] catA (4756, 5014, 6614), [EC: 5.5.1.1] catB (3246, 4194, 5254), [EC: 5.3.3.4] catC (782, 862, 998),

[EC: 3.1.1.24] pcaD (4454, 4412, 13,186), pcaL (166, 330, 1006), [EC: 2.8.3.6] pcaJ (2828, 2676, 8192), pcal (3148, 3606, 9364), [EC: 2.3.1.174] pcaf (72, 276, 2194), and [EC: 2.3.1.16] fadA, fadI (6018, 7826, 9860). Furthermore, the abundance of key functional genes involved in pathway A-2 in all reactors stands as follows: [EC: 1.13.11.2] catE (2112, 3384, 6668), dmpB, xylE (1782, 3924, 4470), [EC: 3.7.1.9] dmpD, xylF (1342, 2708, 3070), [EC: 4.2.1.80] mhpD (3186, 4312, 5186), [EC: 4.1.3.39] bphI, xylK, nahM, tesG (942, 1630, 3214) mhpE (3308, 3310, 2610), and [EC: 1.2.1.10] mhpF (2282, 2798, 2862), adhE (934, 1132, 2010). It indicated that the micro-oxygen environment in MAHA-2 promoted the enrichment of functional enzyme genes in the degradation pathway of phenol, thereby increasing the potential for expressing and producing a greater quantity of functional enzymes, and facilitating phenol degradation in MAHA-2 reactor.

In pathway B, the functional gene abundance comprises [EC:4.1.1.61] bsdC, [EC:6.2.1.27] hbaA (948, 124, 24), [EC:1.1.7.1] hcrC, hbaB (178, 402, 276), hcrA, hbaC (120, 360, 216), hcrB, hbaD (204, 366, 344), [EC:1.3.7.8] bcrC, badD (502, 486, 402), bcrB, badE (620, 402, 464), [EC:4.2.1.100] dch (212, 202, 346), [EC:1.1.1.368] had (370, 304, 386), [EC:3.7.1.21] oah (160, 84, 38). It is evident that the gene abundance of phenol degradation enzymes is generally lower in pathway B. Conversely, the gene abundance in the final period consists of [EC:1.1.1.35] fadJ (3584, 7996, 16,882), fadN (7112, 16,538, 25,996), fadB (910, 586, 1328), [EC:2.3.1.16] fadA, fadl (6018, 7826, 9860), [EC:4.2.1.17] crt (4598, 5054, 9280), paaF, echA (7232, 18,688, 29,868). It can be observed that higher gene abundance exists in the end of reaction, possibly due to the shared terminal point of tricarboxylic acid cycle in anaerobic and micro-oxygen microbial activities. In MAHA-2 reactor, enzyme genes near the end of reaction are significantly higher compared to MAHA-1 and HA reactors, indicating that micro-oxygen condition can promote the overall activity of microbial community, thus enhancing the removal efficiency of pollutants.

4. Conclusions

In this study, the cumulative VFAs concentrations in the effluents of HA, MAHA-1, and MAHA-2 reactors reached 184.2, 189.9, and 221.8 mg/L, respectively. TU value of MAHA-2 reactor (~10) after treatment was lower than that of HA and MAHA-1 reactors. MAHA-2 reactor achieved the lower degree of humification and the reduced concentration of benzene-ring organics. And MAHA-2 reactor also displays a remarkable increase in the concentrations of all functional enzymes and ATP. Under this condition, a significant increase in the abundance of functions related to microbial growth and the metabolism of organic compounds. Especially, higher DO concentration promoted the content of protein in EPS. Ultimately, micro-oxygen condition can promote the overall activity of microbial community, thus enhancing the removal efficiency of pollutants.

CRediT authorship contribution statement

Shengqiang Hei: Data curation, Conceptualization. **Biming Liu:** Investigation, Formal analysis. **Hui Xu:** Resources, Project administration. **Xing Lu:** Supervision, Software. **Xinrui Guo:** Visualization, Validation. **Danyang Li:** Resources, Project administration. **Shuo Zhang:** Validation, Resources. **Xia Huang:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2025.123898.

Data availability

The data that has been used is confidential.

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